

# Battery Bidding and Market Efficiency in Energy Transition

Hunt Allcott      Luming Chen      Julia Park\*

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## Abstract

Utility-scale batteries play an increasingly active role in electricity markets by arbitraging energy across time, yet their efficiency depends critically on market design. This paper studies a prevalent feature of real-time electricity auctions: bid lead time, which requires batteries to commit bids in advance of dispatch. Using comprehensive 15-minute bidding and settlement data for all utility-scale batteries in ERCOT from 2018–2025, we document active dynamic bidding, rapid responses to price shocks, and the importance of recent price information for forecasting. We develop an equilibrium model of electricity markets with dynamic battery bidding and show that longer bid lead times substantially reduce operational efficiency and profits, and dampen complementarity between batteries and renewable energy. Allowing bids to depend on state of charge significantly mitigates these losses. Our results highlight bid lead time as a key determinant of battery performance and market outcomes.

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\*Allcott: Stanford University and NBER, email: [allcott@stanford.edu](mailto:allcott@stanford.edu); Chen: University of Michigan, email: [lumingch@umich.edu](mailto:lumingch@umich.edu); Park: Stanford University, email: [julpark@stanford.edu](mailto:julpark@stanford.edu). We are grateful for comments and suggestions by participants at the Energy Camp, UM/MSU EEE Day, and ISO New England Market Design workshop. Comments are welcome.

# 1 Introduction

Utility-scale batteries have expanded rapidly in capacity in the United States and play a critical role in the energy transition. In wholesale electricity markets, batteries serve dual functions: they store energy to complement intermittent renewable generation and act as energy arbitrageurs that smooth intra-day price fluctuations. Batteries actively participate in real-time energy auctions, charging when prices are low and discharging when prices are high. The efficiency of these bidding strategies depends critically on the information available to forecast future prices and optimally time charging and discharging decisions. Despite growing economic interests in energy storages, limited attention has been paid to how batteries operate in electricity markets, how efficiently they bid as energy arbitrageurs, and how bidding (in)efficiencies affect market equilibrium and the broader energy transition.

This paper investigates a key market design feature of real-time electricity auctions that is particularly relevant to battery bidding: the bid lead time, defined as how far advanced batteries should commit their bids. The bid lead time is a common design in the wholesale electricity market around the world and varies across markets. For example, in California ISO (CAISO), batteries need to submit bids for every 60-minute period, commit 75 minutes before start of period, whereas in MidContinent Independent System Operator (MISO), Southwest Power Pool (SPP), and ISO New England (ISO-NE), batteries need to commit their bids 30 minutes in advance. Bid lead time has also received regulatory attention. The CAISO Market Surveillance Committee, for example, has noted that “CAISO should implement rules that enable battery operators to more efficiently manage their state of charge over the day, such as more frequent offer price or limit changes with shorter time lags, state of charge dependent bids and offers, and/or perhaps other design changes.” The bid lead time affects the efficiency of battery operations, because it limits access to the most recent price information and constrains the flexibility of adjusting the state of charge. We evaluate how bid lead time shaping battery bidding efficiency in the context of Electric Reliability Council of Texas (ERCOT), where rapidly expanding storage capacity and detailed high-frequency bidding data provides an ideal setting to study battery bidding behavior and real-time market designs. Moreover, ERCOT effectively has no meaningful bid lead time in real-time, which provides an ideal setting where batteries update bids using near real-time information.

We compile a comprehensive data set covering all utility-scale batteries in ERCOT from 2018 to 2025, and we access the real-time bidding and settlement information for every 15 minutes. Using these high-frequency data, we document three key data patterns. First, we find that batteries are actively bidding into the real-time energy market, charging during low-price periods and discharging during high-price periods to maximize profits. Batteries submit high generation bid prices to avoid being dispatched too early and to preserve the high option value of stored energy.

As prices approach their peak, batteries sharply lower generation bid prices to ensure dispatch and capture high market prices. Second, we find that batteries respond to unexpected market shocks by quickly adjusting the generation bid prices. This behavior demonstrates that batteries actively exploit the real-time information and flexibly adjust their bidding strategy to accommodate the unexpected changes in market conditions. Third, since accurate prediction of future prices is a key prerequisite for batteries to earn profits, we demonstrate a critical role of recent price information in reducing forecast errors.

Motivated by these empirical facts, we develop an equilibrium model of electricity markets, which embeds a dynamic model of battery bidding. Batteries participate in the real-time market by submitting charge and discharge bids every 15 minutes, which determine whether they demand or supply energy depending on realized prices. Because electricity is storable, batteries face an inherently dynamic problem: discharging this period generates immediate profits but reduces the ability to supply energy in future high-price periods, while charging stores energy for future arbitrage. We model this trade-off as a dynamic optimization problem in which batteries choose bids to maximize expected profits, taking into account stochastic electricity prices and the evolution of their state of charge. A key feature of the model is that bidding behavior encodes the inter-temporal opportunity cost of stored energy. Optimal discharge bids reflect both current price conditions and the expected value of retaining energy for future periods, while optimal charge bids reflect the discounted value of increasing state of charge. In the baseline environment, bids are submitted just before dispatch, allowing batteries to condition on up-to-date information about prices and system conditions. This benchmark corresponds to a frictionless setting in which batteries can respond optimally to real-time shocks.

We then introduce bid lead time as a market design constraint such that bids for a given interval must be submitted several periods in advance. This modification generates three key frictions. First, batteries must form expectations using outdated price information, which increases forecast uncertainty. Second, they no longer observe their actual state of charge at the time of dispatch, but instead must plan based on its expected distribution. Third, they lose the ability to adjust bids in response to newly realized shocks. As a result, bidding becomes a problem of decision-making under imperfect information and commitment, leading to inefficient inter-temporal allocation of energy. We embed this dynamic bidding problem in a market equilibrium framework in which battery dispatch interacts with aggregate supply and demand to determine electricity prices. Using the model, we quantify how bid lead time distorts bidding behavior, reduces arbitrage efficiency, and lowers battery profits. These inefficiencies propagate to the market level, affecting price dynamics, consumer surplus, and the distribution of rents across market participants.

We estimate the model using high-frequency data from ERCOT. We first recover a semi-parametric price process that preserves intraday dynamics and price spikes, and then estimate

battery-specific cycling costs using a simulated method of moments, matching observed charging and discharging behavior. The estimated cycling costs range from roughly \$4 to \$20 per MWh for most batteries, consistent with engineering benchmarks, and are tightly disciplined by observed cycling frequency and price volatility. The model fits key features of battery operations, including intraday charging and discharging patterns.

Using the estimated model, we quantify the impact of bid lead time on behavior and outcomes. Longer lead times lead batteries to submit more conservative and less targeted bids, causing them to discharge too early and miss peak-price periods. This distortion generates substantial profit losses: relative to last-minute bidding, a 30-minute lead time reduces profits by about 17%, while a 75-minute lead time reduces profits by roughly 25%. These effects are amplified by higher price volatility and mitigated by longer-duration batteries, which have greater inter-temporal flexibility.

Next, we explore a commonly discussed regulatory proposal of state-of-charge-dependent bids such that batteries could submit a contingent bid price dependent on the realized state of charge. Due to computation limitations, we evaluate a parsimonious two-point state-of-charge-dependent bids. We find that state-of-charge-dependent bids brings substantial improvement to batteries' operation profits, which mitigate quantity uncertainty and allow contingent plans based on different realizations of state of charge. Under 30-minute bid lead time, state-of-charge-dependent bids improve the average profit by 12.3%, and under 75-minute bid lead time, state-of-charge-dependent bids improve the average profit by 15.8%. Allowing contingent bidding plans significantly lowers the uncertainty towards quantities, and improves batteries' operation profits.

Finally, we embed the model in a market equilibrium framework to assess aggregate impacts. Removing bid lead time lowers average electricity prices, especially in the upper tail, by enabling batteries to more effectively arbitrage peak demand. The resulting gains accrue primarily to consumers and battery operators, while reducing profits for conventional generators. Moreover, our preliminary results indicate that bid lead time weakens complementarity between energy storage and solar energy. Overall, we find that bid timing is a quantitatively important dimension of market design: even relatively short lead times generate meaningful inefficiencies in storage operations and distort equilibrium outcomes in electricity markets.

This paper contributes to the following three strands of literature. First, this is the first economics paper, to our knowledge, to empirically evaluate batteries dynamic bidding behavior, which play a critical role in the equilibrium on the wholesale electricity market. Our paper moves away from common assumptions that batteries are perfect energy arbitrageurs and builds in inefficiency from the regulatory environment such that their operation might be compromised. Our paper thus complements previous work on the economic values of energy storages, including [Butters et al. \(2025\)](#), [Holland et al. \(2024\)](#), [Karaduman \(2021\)](#), and [Lamp and Samano \(2022\)](#). Previous work has also examined various aspects of battery operations, exploring the market power

(Andrés-Cerezo and Fabra, 2023), complementarity with renewable energy (Andrés-Cerezo and Fabra, 2025), congestion benefits (Kirkpatrick, 2020), ancillary services (Reynolds, 2025), and long run equilibrium under renewable portfolio standards (Palley et al., 2024). Our paper exploits the high-frequency dispatching and bidding data and estimates fundamental primitives of battery operations.

Second, this paper investigates the design of wholesale electricity market by extending the analyses to battery bidding. Previous papers have examined strategic bidding (Hortaçsu and Puller, 2008), sequential market (Ito and Reguant, 2016), real-time pricing (Allcott, 2012), bidding sophistication (Hortaçsu et al., 2019), forward commodity markets (Jha and Wolak, 2023). This paper incorporates the bidding behavior of energy storages, highlight the unique dynamic feature, and explore the importance of a common market design feature as well as its solutions.

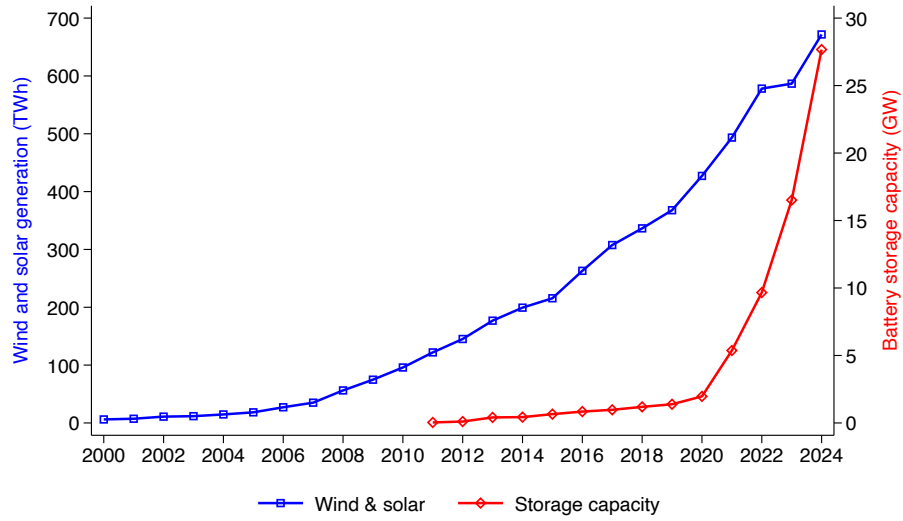
Third, and more broadly, this paper is related to the literature on how auction design shapes information structure and dynamic efficiency, where equilibrium outcomes depend critically on the timing and precision of information available to bidders. The closest related work studies how auction timing and dynamic participation affect efficiency (e.g., Chaves and Ichihashi (2024); Hendricks et al. (2025)). In contrast, we study a different margin of timing: constraints on when bids must be submitted, which restrict the information available at the time of decision-making and generate dynamic inefficiencies even in otherwise well-functioning markets.

## 2 Background and Data

### 2.1 Energy Storages in Energy Transition

Energy storage has become increasingly important in wholesale electricity markets as renewable energy deployment expands. As shown in Figure 1, wind and solar powered energy in the United States has risen from a negligible level in early 2000s to more than 600 TWh (15.5% of total generation) in 2024. In the meantime, utility-scale battery capacity expanded modestly after 2010 and experienced fast growth after 2020. Total battery capacity has exceeded 27 GW in 2024 and grown 13 times from 2 GW in 2020.

**Figure 1: Market Evolution of Renewable Generation and Battery Capacity**



Notes: This figure plots the time trend of wind and solar generation (in blue, y-axis on the left) and the battery capacity (in red, y-axis on the right) in the US from 2000 to 2024.

Batteries play dual roles in wholesale electricity markets. First, they store energy and complement intermittent renewable generation. Because wind and solar output often peak at times that do not coincide with peak electricity demand, batteries can shift energy across time by charging during periods of high renewable generation and discharging during periods of high demand. Second, batteries act as energy arbitrageurs. By charging when prices are low and discharging when prices are high, batteries smooth intra-day price fluctuations.

Our paper focuses on Electric Reliability Council of Texas (ERCOT) which operates the wholesale electricity market in Texas. Texas now hosts the second largest battery capacity in the US, following California. Unlike California, where batteries are often co-located with solar farms and are predominantly four-hour systems, batteries in Texas are mostly stand-alone facilities and are primarily one-hour or two-hour batteries, as shown in Appendix Figures A1 and A2. Batteries in ERCOT participate in both the day-ahead market, primarily to provide ancillary services, and the real-time market for energy arbitrage.<sup>1</sup>

Batteries in ERCOT actively participate in the wholesale auctions every five minutes, with settlements occurring every fifteen minutes. In each interval, a battery submits a *generation* bid

<sup>1</sup> Ancillary services are purchased by ERCOT from the day-ahead market to balance electricity supply and demand next day on the grid and mitigate real time operational issues. Ancillary services in ERCOT include different types: regulation service (regulation up and down), responsive reserve service, non-spin reserve service, and ERCOT contingency reserve service (ECRS). Different types of ancillary services vary across three core dimensions: response speed, direction (inject power or absorb power), and duration. Energy storages participate in all kinds of ancillary services.

to supply energy and a *load* bid to demand energy. These bids are interval- and unit-specific step functions. We illustrate the shape of bid functions for one of the largest batteries in ERCOT, Turquoise Storage, on one hot summer day in 2024, as shown in Appendix Figures A3 and A4. Generation bid prices indicate the minimum price that batteries are willing to supply a certain amount of energy. They are capped from above at \$5,000/MWh, and a generation bid price at the cap could effectively avoid dispatch. Load bid prices indicate the maximum price that batteries are willing to accept to demand a certain amount of energy. They are capped from below at -\$250/MWh, and a load bid price at the floor could effectively avoid charging.

To fully exploit arbitrage opportunities, batteries face a complex dynamic decision of when to charge and when to discharge at high frequency, which requires forecasting future prices. However, increasing penetration of renewable energy exacerbates price volatility, making accurate forecasting more difficult. We measure price volatility using the dispersion between real-time and day-ahead market prices, where the day-ahead price is settled before the real-time market opens and is therefore known to batteries at the time of bidding. We observe a clear increase in this dispersion over time as shown in Appendix Figure A5, indicating growing difficulty in forecasting future prices.

## 2.2 Bid Lead Times in Wholesale Electricity Markets (ISO/RTOs)

The bidding efficiency of batteries depends crucially on the design of the real-time energy auctions, and the wholesale electricity market in general. Obstacles embedded in the market design could potentially hinder efficient bidding of batteries, dampen their operation profitability, and shape the complementarity between energy storages and renewable sources. We focus on one important market design feature: the bid lead time, defined as how far advanced batteries should commit their bids. The bid lead time is a common design in the wholesale electricity market around the world, and it varies across markets. For example, in CAISO, storage resources must submit bids up to 75 minutes prior to the operating hour, after which bids remain fixed for the duration of the hour, whereas in MISO, SPP, and ISO New England, batteries need to commit their bids 30 minutes in advance. In comparison, ERCOT represents a market design with near-zero bid lead time. ERCOT's real-time market features a 5-minute dispatch interval and allows market participants to update energy offer curves at high frequency.<sup>2</sup>

As noted by a recent report on energy storage deployment (GridLab, 2026), long bid lead time is one of the major challenges on the energy market for batteries. Since most wholesale electricity markets require participants to submit bids 30 to 75 minutes prior to the operating hour, while

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<sup>2</sup>Prior to recent reforms, offers were effectively locked several minutes before dispatch through a short Adjustment Period, implying a small but non-zero bid lead time. However, with the implementation of NPRR1058 in 2024, ERCOT allows offer updates during the operating period, effectively eliminating binding bid timing constraints.

market dispatch and settlement occur at much finer temporal resolution (e.g., five-minute intervals), this institutional feature creates a fundamental timing mismatch to batteries whose marginal costs are inherently dynamic. Because the true value of stored energy evolves continuously while bids remain fixed, battery operators must commit to bids that may quickly become outdated, leading to inefficient dispatch and distorted incentives. As emphasized by the report, “this contrast highlights that bid lead times and lock-in requirements are not merely operational details but central design features that shape the efficiency and economic performance of storage participation in wholesale markets.”

Bid lead time has also received regulatory attention. The CAISO Market Surveillance Committee has noted that “CAISO should implement rules that enable battery operators to more efficiently manage their state of charge over the day, such as more frequent offer price or limit changes with shorter time lags, state of charge dependent bids and offers, and/or perhaps other design changes.” As emphasized by the Department of Market Monitoring of CAISO, these bids are formed based on expectations of the resource’s current and future state of charge, yet actual dispatch occurs after this submission, potentially leading to substantial deviations between expected and realized state of charge. Because of this timing mismatch, that storage operators must commit to opportunity-cost-based bids under incomplete information, while their physical state continues to evolve between bid submission and dispatch. Stakeholder comments reinforce the implications of this design. Terra-Gen, one of the major energy storage owners and operators, highlighted that storage operators cannot adjust bids closer to real time, even though realized market outcomes and thus state of charge availability depend on uncertain dispatch conditions beyond the operator’s control.<sup>3</sup> This can result in outcomes that are not fully aligned with the batteries’ actual operating constraints, motivating the need for uplift payments to compensate for inefficient dispatch outcomes induced by market design limitations.

Taken together, the bid lead time as a common market design feature across many ISO/RTOs, imposes a commitment constraint with information lag, where storage resources must make forward-looking bidding decisions under uncertainty while facing limited ability to adjust to realized conditions. This generates a wedge between bid-based optimization and physical feasibility which could have important implications for market efficiency.

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<sup>3</sup>This is quoted from the [CAISO shareholder meeting](#).

## 2.3 Data

We compile a comprehensive dataset covering all 338 utility-scale batteries (Energy Storage Resources, or ESRs) registered in ERCOT from 2018 to 2025.<sup>4</sup> We obtain bidding and settlement data from ERCOT’s publicly available market data, which covers both the day-ahead market (DAM), including ancillary services, and the real-time market (RTM).<sup>5</sup> For each battery in each 15-minute market interval, we observe generation offer curves (submitted to supply electricity), load bid curves (submitted to demand electricity), bid awards, and settlement point prices. Prices in ERCOT are location-specific: each ESR is assigned to a settlement point, and we match all observations to the corresponding real-time settlement point price.

In ERCOT’s real-time market, the dispatch algorithm runs every five minutes and settles dispatch at 15-minute intervals. Batteries participate as two distinct resource types: as a generation resource when discharging, and as a load resource when charging. Each resource submits a separate step-function offer curve per 15-minute delivery interval. The generation offer curve consists of up to 35 price-MW pairs,  $\{(q_k^g, p_k^g)\}_{k=1}^K$ , where  $q_k^g$  is the offered level of output (in MW) when the settlement point price falls in the range  $[p_k^g, p_{k+1}^g)$ , with  $p_{K+1}^g$  set to the system-wide cap of \$5,000/MWh. An offer price at the cap effectively prevents dispatch. The load bid curve contains up to 10 price-MW pairs,  $\{(q_k^l, p_k^l)\}_{k=1}^K$ , where  $q_k^l$  is the bid level of consumption (in MW) when the settlement point price falls in the range  $(p_{k-1}^l, p_k^l]$ , with  $p_0^l$  set to the floor of -\$250/MWh. A load bid at the floor effectively prevents charging, while a bid at the ceiling ensures charging regardless of price. These bounds are defined by ERCOT’s market rules and apply to all resource types.

To summarize bidding behavior into a scalar measure, which we use in our descriptive evidence below, we compute a MW-weighted average bid price for each resource-interval observation. For the generation offer, the average price is

$$\bar{p}^g = \frac{\sum_k p_k^g \cdot (q_k^g - q_{k-1}^g)}{\sum_k (q_k^g - q_{k-1}^g)},$$

with the load bid average computed analogously. This measure weights each price step by its MW coverage, capturing the price of a representative megawatt across the full offer curve.

A practical challenge is that batteries routinely submit extreme prices that reflect dispatch preferences rather than economic valuations. On the generation side, prices near the \$5,000/MWh cap signal a preference not to discharge; on the load side, prices near the -\$250/MWh floor signal a

<sup>4</sup>Our sample ends on December 4, 2025 because ERCOT’s Real-Time Co-Optimization + Batteries (RTC+B) market redesign took effect on December 5, 2025, fundamentally altering how batteries bid and settle in the market.

<sup>5</sup>The data is mainly compiled from 60-Day Security Constrained Economic Dispatch (SCED) Disclosure files, published on a rolling 60-day delay following each delivery date.

preference not to charge, while prices near the ceiling signal unconditional willingness to charge. Since these extremes are governed by ERCOT’s policy bounds rather than by marginal costs, including them directly in the weighted average would conflate strategic on/off signals with genuine price information. We therefore apply two cleaning steps before computing averages. For generation offers, we truncate prices above \$1,000/MWh to missing, treating bids above this threshold as dispatch-avoidance signals, and winsorize prices below -\$100/MWh to -\$100, preserving these as genuine willingness-to-dispatch signals while preventing their scale from distorting the average. For load bids, the logic is symmetric: prices below -\$100/MWh are truncated to missing (charging-avoidance signals), and prices above \$1,000/MWh are winsorized to \$1,000.

ERCOT’s day-ahead market differentiates between two types of participation with distinct implications for real-time operations. For energy offers, the day-ahead market functions as a financial-forward market: a battery awarded energy in the day-ahead market receives the day-ahead settlement point price for the awarded quantity but incurs no physical dispatch obligation; actual dispatch is determined independently by the real-time market based on the battery’s real-time offer curves, with any imbalance between day-ahead-awarded and actual real-time quantities settled at the real-time price. Because day-ahead energy participation is purely financial and does not constrain real-time behavior, we make limited use of it in the analysis of this paper.

By contrast, ancillary service awards from the day-ahead market operate differently and are physically binding. A battery awarded, for instance, Regulation Up capacity in the day-ahead market must hold that capacity available for deployment in real-time, directly constraining the energy available for arbitrage. We take ancillary service participation into account when characterizing the data, but our main analysis focuses on batteries that participate exclusively in the real-time energy market.

We complement the bidding data with battery characteristics from the U.S. Energy Information Administration (EIA) Form EIA-860. This annual survey provides plant-level information on installed power capacity (MW), energy capacity (MWh), commercial operation date, and geographic location.<sup>6</sup> We match ERCOT resource names to EIA plant identifiers using a combination of text-based name matching and manual verification. These data allow us to characterize the variation in battery size and duration within our sample, and to normalize bid quantities by capacity where needed.

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<sup>6</sup>Energy capacity (MWh) is installed power capacity (MW) multiplied by battery duration.

### 3 Descriptive Evidence

#### Fact 1: Batteries bid dynamically and option value matters

We document dynamic battery bidding patterns in Figure 2. For each 15-minute interval of the day, we compute average generation and load bid prices. ERCOT real-time prices exhibit a clear bimodal pattern, with a modest morning peak around 7 a.m. and a much larger evening peak around 7 p.m., both coinciding with periods of high electricity demand. Prices are lowest from the morning through early afternoon (approximately 8 a.m. to 2 p.m.), reflecting the abundance of renewable generation. In response to this intraday price variation, batteries adjust both their generation and load bid prices dynamically. Generation bid prices reflect batteries' willingness to sell energy, and higher generation bid prices reduce the likelihood of dispatch. Conversely, load bid prices capture batteries' willingness to buy energy and higher load bid prices increase the likelihood of being charged.

As shown in Panel (a), batteries submit relatively high generation bid prices several hours before the peak period to avoid being discharged too early, reflecting the high option value of stored energy. It is more profitable to retain energy for later, higher-price hours than to sell during periods of relatively low prices. As the price peak approaches, generation bid prices drop sharply to ensure dispatch, allowing batteries to sell when prices are highest. We observe a similar pattern around the morning peak, although the magnitude is more modest.

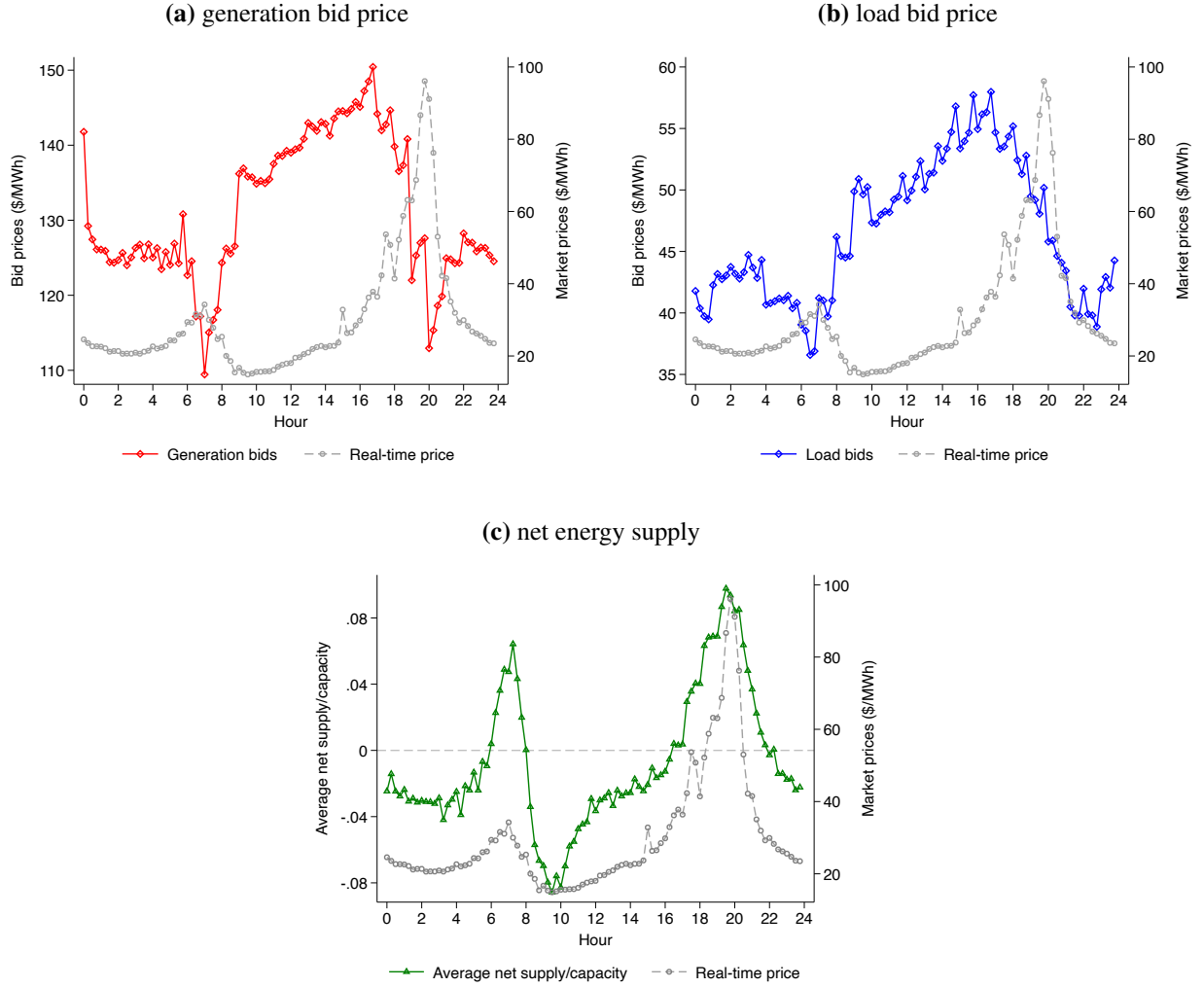
As shown in Panel (b), we find that load bid prices exhibit a comparable intraday pattern, albeit at lower levels. Batteries submit relatively high load bid prices several hours before the peak period, again reflecting the high option value of stored energy and the incentive to secure full charge ahead of high-price periods. As peak hours arrive, load bid prices decline sharply, as batteries avoid charging when grid prices are elevated. As with generation bids, the same pattern for load bid prices appears around the morning peak, though with smaller magnitudes.

Consistent with the bidding strategy, the average net supply, defined as the difference between the total discharged and charged energy, closely track the price pattern, as shown in Panel (c). Batteries are net energy suppliers during morning and evening peak hours, and net energy consumers during the rest of the day. Overall, these patterns reveal a clear dynamic bidding behavior of batteries. They are actively bidding into the real-time energy market to charge during low-price periods and discharge during high-price periods to maximize profits. The intra-day bid price patterns follow closely the option values of stored energy.<sup>7</sup>

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<sup>7</sup>Most batteries in ERCOT have adopted automated bidding tools to allow real-time responses of bid functions to market information and forecast. See early report on automated bidding by [Utility Drive](#) and a [report](#) from Key Capture, one of the largest battery owners in ERCOT. We discuss one concrete example of automated bidding adoption in Appendix. We find that most recently installed batteries have not engaged in manual bidding since coming online.

**Figure 2: Bid Prices and Net Energy Supply of Batteries**



Notes: This figure plots the average generation bid prices (red line in Panel (a)), load bid prices (blue line in Panel (b)), and the net energy supply as a share of capacity (green line in Panel (c)) in 2024. We truncate both generation and load bid prices at an upper bound of \$1,000/MWh and winsorize them at a lower bound of -\$100/MWh. Real-time market prices are plotted in grey with y-axis on the right for each panel.

## Fact 2: Batteries respond to unexpected market shocks

We examine how batteries respond to unexpected market shocks. We identify shocks at the settlement-point-by-15-minute-interval level by impose two restrictions. First, shocks must be associated with unplanned generation or transmission outages lasting more than two hours.<sup>8</sup> Second,

<sup>8</sup>A settlement point is defined as the location where prices are calculated for settlement, including Resource Nodes, Load Zones, and Hubs, according to [official protocols](#).

we require that these outages generate abrupt and substantial changes in market conditions. Specifically, we classify an interval as a shock if, following the onset of the outage, the real-time price exceeds the day-ahead price by at least \$100/MWh, while the real-time–day-ahead price spread remains within  $[-50, 50]$  \$/MWh over the preceding three hours. This criterion isolates episodes in which real-time prices deviate sharply from day-ahead expectations despite relatively stable conditions immediately beforehand, capturing genuinely unanticipated disruptions due to outages rather than gradual market adjustments.

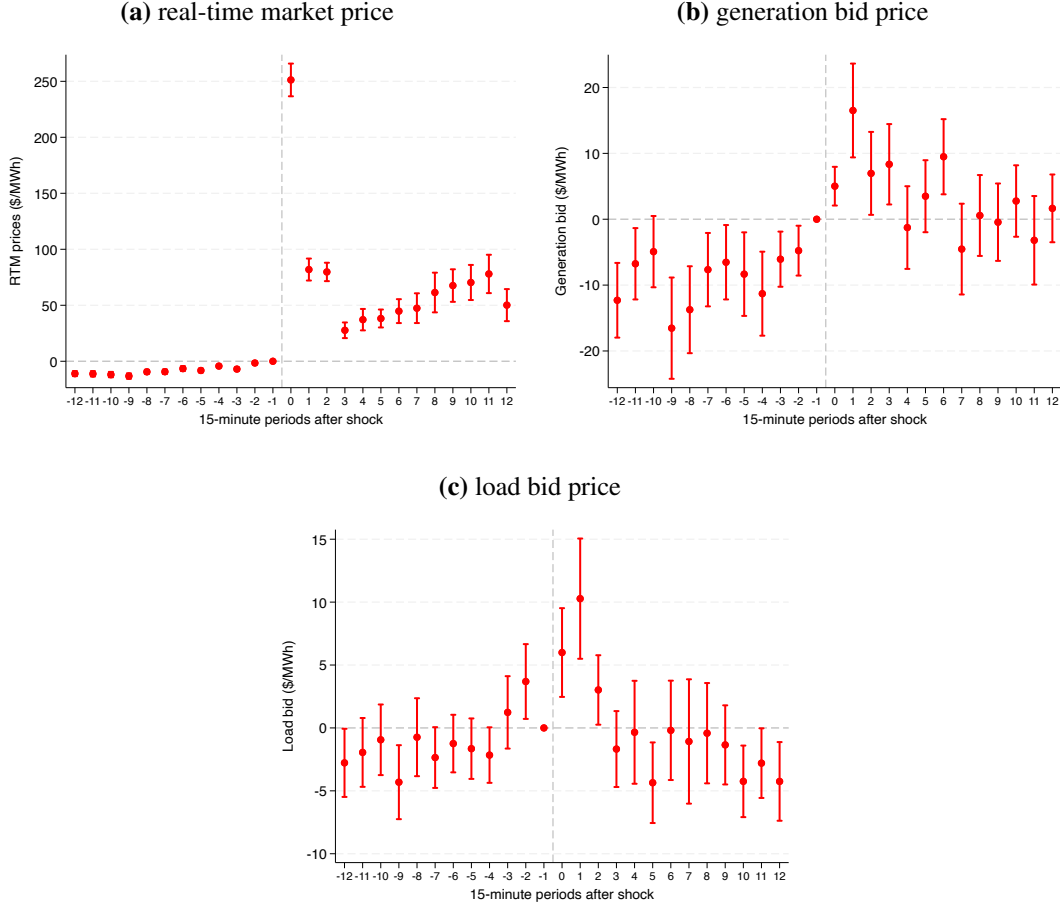
We estimate the following event-study specification to evaluate how batteries respond to unexpected market shocks.

$$y_{it} = \sum_{k=-12, k \neq -1}^{12} \beta_k \mathbb{1}(T_{it} = k) + \alpha Price_{it}^{DAM} + X_{it} + \epsilon_{it}. \quad (1)$$

We index battery by  $i$  and 15-minute intervals by  $t$ . The dependent variable  $y_{it}$  denotes key outcome variables, including market prices and bid prices.  $T_{it}$  represents event time relative to the start of a market shock, with  $k = -1$  omitted as the reference period. Our key coefficients of interest is  $\beta_k$ , which measures how unexpected market shocks affect the dependent variable  $k$  intervals relative to the onset of the shock. We control for day-ahead market prices  $Price_{it}^{DAM}$ , and include a rich set of fixed effects  $X_{it}$ , consisting of battery fixed effects interacted with month and hour dummies. These fixed effects absorb battery characteristics, seasonality in electricity demand, intra-day volatility, as well as variations in generation resource mixes around each battery. The key identification assumption is that, conditional on the rich controls of day-ahead market prices and unobserved shocks at the battery, month, and hour levels, the timing of market shocks is exogenous to individual batteries.

The event-study estimation results are presented in Figure 3. We confirm that our identified unexpected market shocks drive price spikes in Panel (a), as we observe an abrupt rise in real-time market price at the onset of the shock, and the increase in the price level lasts for more than three hours. On the contrary, we observe a stable level of the real-time market price before the event. More importantly, we find that batteries respond to such shocks by increasing the generation and load bid price levels. Batteries may anticipate persistence in price shocks and therefore optimally respond by increasing both generation and load bids, allowing them to cycle multiple times during periods of elevated prices. This behavior demonstrates that batteries actively exploit the real-time information and flexibly adjust their bidding strategy to accommodate the unexpected changes in market conditions.

**Figure 3: The Impacts of Unexpected Market Shocks**



Notes: This figure plots the estimation results of equation (1). Panel (a) plots the impacts of unexpected market shocks on real-time prices, panel (b) plots the coefficients on generation bid prices, and panel (c) plots the coefficients on load bid prices. 95% confidence intervals are constructed from clustered standard errors at the battery level.

### Fact 3: Recent prices are most informative for price projections

We investigate the role of recent price information in shaping expectations about future prices. Using real-time market prices at a 5-minute frequency, we estimate the following empirical specification to evaluate how much variation in current prices can be explained by past price realizations.

$$\log p_{d,h} = \gamma_h \log p_{d,h-\tau} + \phi \text{Price}_{d,h}^{\text{DAM}} + X_{d,h} + \epsilon_{d,h}. \quad (2)$$

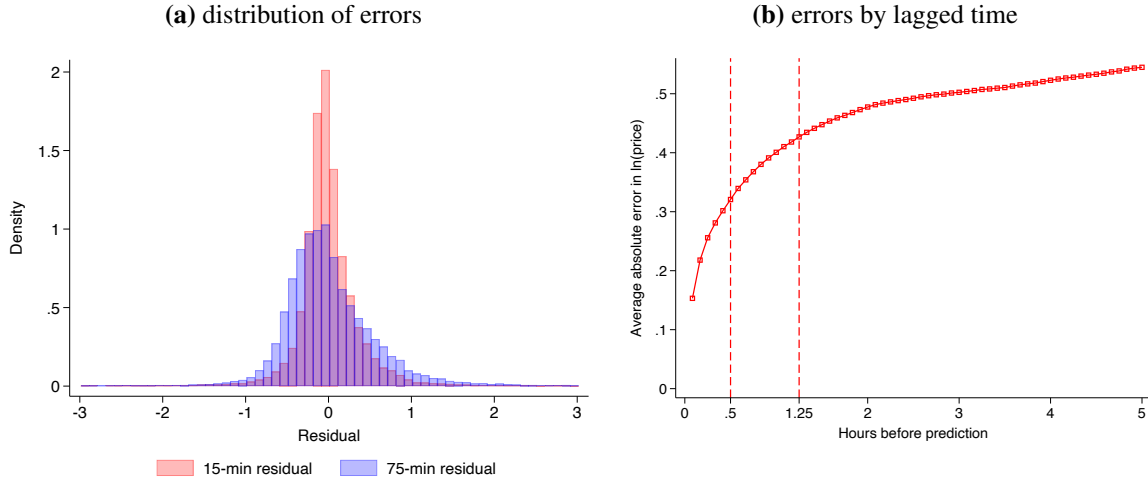
Here,  $d$  denotes the day and  $h$  indexes the intra-day 5-minute interval. The dependent variable is the log real-time market price at day  $d$  and interval  $h$ , and the key independent variable is lagged

log real-time prices  $\tau$  intervals earlier. We allow the coefficients on lagged prices to vary flexibly across intra-day intervals. Moreover, we control for hourly day-ahead market prices  $Price_{d,h}^{DAM}$ , and include month fixed effects to absorb seasonal variations in prices.

The main results are summarized in Figure 4. Panel (a) plots the distribution of prediction residuals  $\hat{\epsilon}_{d,h}$  under two specifications: one using only a 15-minute lag ( $\log p_{d,h-3}$ ) to form a 15-minute-ahead forecast, and another using only a 75-minute lag ( $\log p_{d,h-15}$ ) to form a 75-minute-ahead forecast. Extending the forecasting horizon from 15 to 75 minutes substantially increases the dispersion of prediction errors, with the standard deviation rising by 43.9%.

Panel (b) plots the average absolute prediction errors as a function of forecasting horizon. We observe a sharp increase in errors, particularly within the first two hours. Overall, these results highlight the importance of recent price information for accurately predicting future real-time prices.

**Figure 4: Projection Errors of Prices**



Notes: This figure plots the estimation results of equation (2). Panel (a) plots the distribution of forecast errors when  $\tau=3$  (a 15-minute-ahead prediction) and when  $\tau=15$  (a 75-minute-ahead prediction), while panel (b) plots the average absolute prediction errors as a function of forecasting horizon.

Taking stock, these empirical patterns document batteries' dynamic bidding incentives and illustrate how batteries adjust bid prices to exploit intra-day price volatility to maximize profits. Moreover, we also show that batteries respond rapidly to unexpected market shocks in the real-time market. Accurate prediction of future prices is a key prerequisite for earning profits, and we document the critical role of recent price information in reducing forecast errors. These findings

motivate our structural model, in which we characterize batteries' dynamic bidding behavior in the real-time market, and analyze how batteries optimally adjust their bids in the presence of bid lead time that restricts the effective prediction horizon.

## 4 A Structural Model of Battery Bidding in Wholesale Market

### 4.1 Dynamic Bidding of Batteries

We index batteries by  $i$ , each with energy capacity  $k_i$ , defined as the maximum amount of energy the battery  $i$  can discharge. Time  $t$  is discretized into 15-minute intervals. We index days by  $d$  and intra-day intervals by  $h \in \{1, 2, 3, \dots, 96\}$ . At the beginning of each interval  $t$ , battery  $i$  has a remaining state of charge  $s_{it} \in [0, 1]$ , defined as the level of charge relative to its capacity. Battery  $i$  submits a load bid to get charged and a generation bid to get discharged and those bids are interval- and unit-specific step functions. For tractability, we assume that batteries submit a single generation bid price  $b_{it}^g$  and load bid price  $b_{it}^l$  with exogenous and constant bid quantities  $q_i^g$  and  $q_i^l$  for each interval.  $b_{it}^g$  represents the generation bids such that the battery is willing to *supply* power  $q_i^g$  when the real-time market price is above  $b_{it}^g$ . Similarly,  $b_{it}^l$  represents the load bids such that the battery is willing to *buy* power  $q_i^l$  when the real-time market price is below  $b_{it}^l$ .

We denote the real-time market price in interval  $h$  as  $p_h$ , which follows a Markov process  $F_h(p_h|p_{h-1})$ . Specifically, we assume that

$$p_t = \phi_h p_{t-1} + \xi_h + \epsilon_t. \quad (3)$$

Coefficients  $\phi_h$  and  $\xi_h$  denote the AR(1) slope and intercept specific to each intra-day interval, respectively. For the rest of illustration of the model, we focus on batteries' bidding problem within a specific day, and we therefore substitute all time stamps  $t$  with intra-day interval  $h$ . For ease of exposition, we abbreviate  $F_h(p_h|p_{h-1})$  as  $F_h(p_h)$ .

We denote round-trip efficiency  $\delta < 1$ , defined as the ratio of discharged energy to charged energy and measures the efficiency loss in the cycle. Since batteries are subject to the physical constraints such that they can only discharge until the remaining state of charge  $s_h$  drops to 0, and can only charge up to full state of charge such that  $s_h = 1$ , the bidding quantities can be defined as follows.

$$q_h^g = q_i^g \times \mathbb{1}(s_h > 0), \quad q_h^l = \frac{q_i^l}{\delta} \times \mathbb{1}(s_h < 1).$$

The battery will be dispatched to charge if the real-time market price  $p_h$  is below the load bid price  $b_h^l$ , and discharge if real-time market price  $p_h$  is above the generation bid price  $b_h^g$ . Consequently, the discharged energy  $a_h$  and its probability distribution of which are shown as below.

$$a_h = q_h^g \mathbb{1}(p_h \geq b_h^g) - q_h^l \mathbb{1}(b_h^l \geq p_h), \quad P_h(a_h) = \begin{cases} 1 - F_h(b_h^g), & a_h = q_h^g \\ F_h(b_h^l), & a_h = -q_h^l \\ F_h(b_h^g) - F_h(b_h^l), & a_h = 0 \end{cases}$$

The battery is subject to the cycling cost as  $\zeta \times |a_h|$ , where  $\zeta$  denotes the cycling cost per unit of energy charged/discharged.<sup>9</sup> Therefore, the expected profit function is defined as follows, which is the difference between revenues from discharged energy and cost from charged energy, minus the cycling cost.

$$\begin{aligned} \mathbb{E}_h[\pi(\mathcal{B}_h)] &= \mathbb{E}_h[a_h \times p_h - \zeta \times |a_h|] \\ &= \mathbb{E}_h\{[q_h^g \mathbb{1}(p_h \geq b_h^g) - q_h^l \mathbb{1}(b_h^l \geq p_h)] \times p_h - \zeta \times [q_h^g \mathbb{1}(p_h \geq b_h^g) + q_h^l \mathbb{1}(b_h^l \geq p_h)]\} \\ &= \underbrace{\int_{p_h \geq b_h^g} q_h^g p_h dF_h(p_h)}_{\text{revenues from discharging}} - \underbrace{\int_{p_h \leq b_h^l} q_h^l p_h dF_h(p_h)}_{\text{costs from charging}} - \underbrace{\zeta \times [q_h^g (1 - F_h(b_h^g)) + q_h^l F_h(b_h^l)]}_{\text{cycling cost}}. \quad (4) \end{aligned}$$

The expected transition dynamics of  $s_h$  is defined as follows, together with the probability distribution of  $s_{h+1}$ .

$$s_{h+1} = s_h - q_h^g \mathbb{1}(p_h \geq b_h^g) + \delta q_h^l \mathbb{1}(b_h^l \geq p_h), \quad P_h(s_{h+1}) = \begin{cases} 1 - F_h(b_h^g), & s_{h+1} = s_h - q_h^g \\ F_h(b_h^l), & s_{h+1} = s_h + \delta q_h^l \\ F_h(b_h^g) - F_h(b_h^l), & s_{h+1} = s_h \end{cases}$$

A battery decides when to charge or discharge for every interval  $h$  throughout the day, with the dynamic optimization implemented through its submitted bids. By submitting a lower generation bid price in the current period, batteries are more likely to be dispatched and earn a profit from discharging energy. However, discharging reduces the state of charge, making the battery less able to continue supplying energy in subsequent periods without recharging first. The value function can be formulated as follows with a discount factor  $\beta$ . If the price is expected to be high in the current period, batteries tend to submit a low generation bid price  $b_h^g$  to guarantee being discharged. If the price is expected to be high in the future periods, batteries tend to submit a high generation bid price  $b_h^g$  to avoid being discharged early and might submit a high load bid price  $b_h^l$  to ensure

<sup>9</sup>This functional form is a one-segment special case of [Xu et al. \(2017\)](#).

recharging. This trade-off is more obvious from the expression of the continuation value in equation (6), as a lower generation bid price  $b_h^g$  increases the profit from the current period, but reduces the continuation value by moving the value function to low SoC state.

$$V_h(s_h, p_{h-1}) = \max_{b_h^g, b_h^l} \left\{ \underbrace{\mathbb{E}_h[\pi(b_h^g, b_h^l)]}_{\text{profit from the current period}} + \underbrace{\beta \mathbb{E}_h[V_{h+1}(s_{h+1}, p_h) | s_h, p_{h-1}]}_{\text{continuation value from the future}} \right\} \quad (5)$$

$$\begin{aligned} \mathbb{E}_h[V_{h+1}(s_{h+1}, p_h) | s_h, p_{h-1}] = & \int_{p_h} \{ [1 - F_h(b_h^g)] V_{h+1}(s_h - q_h^g, p_h) + [F_h(b_h^g) - F_h(b_h^l)] V_{h+1}(s_h, p_h) \\ & + F_h(b_h^l) V_{h+1}(s_h + \delta q_h^l, p_h) \} dF_h(p_h | p_{h-1}). \end{aligned} \quad (6)$$

We solve for the optimal generation bid price  $b_h^g$  and load bid price  $b_h^l$  as follows.

$$b_h^g = \zeta + \beta \times \frac{\int [V_{h+1}(s_h, p_h) - V_{h+1}(s_h - q_h^g, p_h)] dF_h(p_h | p_{h-1})}{q_h^g}. \quad (7)$$

$$b_h^l = -\zeta + \beta \times \frac{\int [V_{h+1}(s_h + \delta q_h^l, p_h) - V_{h+1}(s_h, p_h)] dF_h(p_h | p_{h-1})}{q_h^l}. \quad (8)$$

The optimal bid prices have intuitive interpretations. The optimal generation bid price measures the willingness to discharge for batteries and covers the opportunity cost of discharging early. The optimal load bid price measures the willingness to charge for batteries and captures the average discounted value of extra state of charge in the next period. Cycling cost will make bids more conservative and be less likely to result in cycling.

## 4.2 Bidding under Lead Time

We introduce bid lead time into the problem: bids for period  $h$  must be submitted  $z$  periods in advance, at the beginning of  $h - z$ . Relative to the baseline case of last-minute bidding, this modification introduces three key changes to the battery's dynamic optimization problem. First, instead of observing most recent price information when forming price expectations, the battery only observes the information available at time  $h - z$  and must form expectations  $z$  periods ahead. Second, the battery no longer observes the state of charge at the beginning of the target period  $s_h$ . Instead, it observes  $s_{h-z}$  and must form expectations over the distribution of  $s_h$ . Third, lead time reduces operational flexibility. In particular, bids can no longer respond to price shocks realized

last period, as they have already been committed  $z$  periods in advance.

With a lead time  $z$ , batteries need to formulate expectation over  $p_h$  conditional on its current price information available at time  $h-z$ ,  $p_{h-z-1}$ . We denote the conditional distribution of  $p_h$  given  $p_{h-z-1}$  as  $G_h^z(p_h | p_{h-z-1})$ , abbreviated as  $G_h^z(p_h)$ . The distribution of  $G_h^z(p_h)$  evolves recursively according to the underlying one-period price transition kernels  $F_{h-k}(\cdot | \cdot)$ .

$$G_h^z(p_h | p_{h-z-1}) = \int \cdots \int F_h(p_h | p_{h-1}) \prod_{j=1}^z dF_{h-j}(p_{h-j} | p_{h-j-1}), \quad G_h^0(p_h | p_{h-1}) = F_h(p_h | p_{h-1}).$$

The expected  $z$ -period transition dynamics of  $s_h$  is a more complicated object. In each period, the state can evolve in one of three ways: discharge, charge, or remain unchanged, and thus there would be  $3^z$  possible trajectories during  $z$  periods. Those trajectories depend on the current state-of-charge  $s_{h-z}$ , price evolution  $F_h$ , as well as all interim bid prices that have already been committed, denoted as  $\mathcal{B}_{h-z}$ .

$$\mathcal{B}_{h-z} = \{b_{h-z}^g, b_{h-z}^l, b_{h-z+1}^g, b_{h-z+1}^l, b_{h-z+2}^g, b_{h-z+2}^l, \dots, b_{h-1}^g, b_{h-1}^l\}.$$

The expected  $z$ -period transition dynamics of  $s_h$  can be written as follows.

$$P_h^z(s_h | s_{h-z}, \mathcal{B}_{h-z}) = \sum_{s_{h-1}} \cdots \sum_{s_{h-z+2}} \sum_{s_{h-z+1}} \prod_{j=-z}^{-1} P_{h+j}(s_{h+j+1} | s_{h+j}, b_{h+j}^g, b_{h+j}^l). \quad (9)$$

Therefore, the profit function under bid lead time can be written as follows. Compared to profit functions under last-minute bidding in equation (4), the expected profit under bid lead time requires  $z$ -period projection of both prices and state of charge.

$$\begin{aligned} & \mathbb{E}_h[\pi(b_h^g, b_h^l, s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z})] \\ &= \sum_{s_h} \left\{ \int_{p_h \geq b_h^g} q_h^g \times p_h dG_h^z(p_h) - \int_{p_h \leq b_h^l} q_h^l \times p_h dG_h^z(p_h) - \zeta \times [q_h^g(1 - G_h^z(b_h^g)) + q_h^l G_h^z(b_h^l)] \right\} \\ & \times P_h^z(s_h | s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z}). \end{aligned} \quad (10)$$

In period  $h-z$ , the battery bid for interval  $h$ , conditional on its current state-of-charge  $s_{h-z}$ , previous-period price  $p_{h-z-1}$ , as well as all interim bid prices that are pre-committed  $\mathcal{B}_{h-z}$ . The

value function can be formulated as below.

$$\begin{aligned}
V_{h-z}(s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z}) = & \max_{b_h^g, b_h^l} \{ \mathbb{E}_h[\pi(b_h^g, b_h^l, s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z})] \\
& + \beta \mathbb{E}_h[V_{h-z+1}(s_{h-z+1}, p_{h-z}, \mathcal{B}_{h-z+1}) | b_h^g, b_h^l, s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z}] \}.
\end{aligned} \tag{11}$$

### 4.3 Market Equilibrium

Given batteries' optimal bids from their dynamic bidding process as illustrated in Section 4.1 and 4.2, we consider the market equilibrium with batteries actively bidding and getting dispatched. We denote total electricity demand in period  $t$  as  $D_t(p_t)$ , and total electricity generation from renewable energy as  $R_t$ . Therefore, the net demand  $X_t$  is defined as the difference between the total demand and renewable generation such that  $X_t(p_t) = D_t(p_t) - R_t$ . The net demand  $X_t$  consists of two components: the net demand comes from the net charge of batteries  $X_t^B$ , and other demand sources  $X_t^{NB}$ .

The net electricity demand is met by other non-renewable generation resources. We use  $S_t(p_t)$  to represent the aggregate bid offer curve across non-renewable generators, and the electricity market clears following the condition below, which yield equilibrium price  $p_t^*$ .

$$X_t(p_t^*) = X_t^B(p_t^*) + X_t^{NB} = S_t(p_t^*). \tag{12}$$

Therefore, the market equilibrium where batteries are allowed to bid last-minute can be defined as follows.

1. Battery  $i$  solves its dynamic bidding problem and submit bids according to equations (7) and (8), given the expected price process as in equation (3).
2. The aggregated net charge from batteries, together with non-battery electricity market, net of electricity generation from renewable energy, are supplied by non-renewable generation resources. This determines the market price according to equation (12).
3. Batteries have rational expectation of price process and therefore equation (3) holds true in equilibrium.

The market equilibrium under bid lead time can be formulated similarly, except that the optimal bids are solved from the dynamic optimization problem under bid lead time. Our equilibrium definition is similar to a moment-based Markov Equilibrium (Ifrah and Weintraub, 2017), where the

key aggregate moments tracked by each battery in the dynamic bidding problem is the equilibrium electricity prices.<sup>10</sup>

## 4.4 Model Discussion

Our model captures the dynamic bidding incentives of batteries as well as equilibrium responses of electricity prices. However, for the tractability of the model, we make several simplification assumptions, and we briefly discuss the implications and empirical justifications of these assumptions.

**Simplified bidding space** We assume that batteries submit a single generation bid price and a single load bid price for each 15-minute period, although in practice they can submit multi-unit bid functions with more than ten segments.<sup>11</sup> To characterize typical bidding behavior, we analyze battery bids in 2024. We define a bid segment as *valid* if its price lies between -\$100 and \$1,000/MWh, as the probability of prices falling outside this range is approximately 0.1% in either tail. In addition, we merge adjacent segments whose bid quantities differ by less than 0.1 MW or whose bid prices differ by less than \$0.1/MWh. Appendix Figure A7 plots the distribution of valid bid segments across batteries and 15-minute intervals. We find that 84% of generation bids contain no more than three valid segments. Tracking a single generation bid price and a single load bid price makes the analysis of bid lead times feasible by avoiding a substantially higher-dimensional state space.

Moreover, we assume that bid quantities  $q_i^g$  and  $q_i^l$  are exogenous and constant within each interval, although the data exhibit variation in quantities across periods. We decompose the variations in bid quantities in Appendix Table A14. Battery fixed effects alone explain more than 80% of variation in bid quantities across batteries and 15-minute intervals. Adding battery-by-day or battery-by-time-of-day fixed effects further improves model fit, but only modestly. In contrast, the time variation alone, captured by day-by-interval fixed effects, explains only about 2% of the total variation. These results suggest that most of the variation in bid quantities is driven by cross-battery differences. Accordingly, our assumption of time-invariant quantities abstracts from relatively smaller temporal variation, while simplifying the state space and facilitating the analysis of bid lead times.

**Abstract from ancillary service market** Batteries in ERCOT also actively participate in the day-ahead ancillary service market. Because our primary focus, the bid lead time, is a feature

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<sup>10</sup>Several recent empirical work using similar equilibrium definitions, such as Barwick et al. (2025), Garcia-Osipenko et al. (2025), Chen (2025), among others.

<sup>11</sup>In the data, however, no battery submitted a 15-minute bid function with more than 12 segments in 2024.

specific to the real-time market, we abstract from batteries’ participation in ancillary services in the model. Accordingly, we restrict the estimation sample to batteries that exclusively participate in the real-time market. Moreover, Appendix Figure A8 shows that as ancillary service markets become saturated, batteries increasingly rely on revenues from the real-time energy arbitrage, with the share rising from less than 10% in 2020 to over 50% in 2025. Given our focus on the real-time market, our simulations are most relevant for a near-future grid in which ancillary service markets are saturated and batteries derive the majority of their revenues from real-time arbitrage, consistent with Butters et al. (2025). Recent work on ancillary services by Reynolds (2025) complements our analysis of battery operations and bidding behavior.

**Ramping of fossil-fueled generators** We endogenize real-time electricity prices in the market equilibrium by allowing battery net discharge to shift total quantity demanded along the electricity supply curve. However, we assume that the electricity supply curve does not respond to changes in battery bidding behavior. This assumption reflects our focus on short-run equilibria, where we compare alternative market designs for bidding while holding the composition of generation technologies fixed. As shown in Appendix Figure A9, total electricity generation in ERCOT from coal and gas plants reached 360,000 GWh in 2024, whereas total battery capacity is only about 8 GW. Consequently, changes in bidding behavior within the existing battery fleet are unlikely to meaningfully affect ramping decisions or induce generation mix changes.

**Additional price predictors** Batteries may, in principle, exploit a rich set of information to form price forecasts and guide bidding decisions. In contrast, our model adopts a parsimonious state space in which batteries use only the most recent price realization to forecast future prices, and combine this information with their current state of charge to determine optimal bids. Importantly, as we show below, the introduction of bid lead time rapidly expands the effective state space even when starting from this minimal set of state variables under last-minute bidding.

Our primary counterfactual comparison is between last-minute bidding and alternative regimes with positive bid lead times. While batteries also observe additional predictors such as day-ahead prices and renewable generation forecasts, these variables remain fully observable at the time bids are submitted, even under bid lead time. As a result, the most recent real-time price effectively summarizes the relevant information set and continues to update dynamically. To assess the empirical validity of this reduced-form information structure, we compare forecast errors from equation (3) in the data and in model simulations (see Appendix Figure A10). Overall, the model closely replicates the distribution of forecast errors observed in the data.

## 5 Estimation and Simulation

### 5.1 Empirical price process

We estimate the electricity price process directly from the observed data as a semi-parametric model that preserves the empirical distribution and intra-day dependence structure of error terms.

$$p_t = \phi_h p_{t-1} + \xi_h + \epsilon_{d,h}$$

We use  $\epsilon_{d,h}$  to denote the residual at period  $h$  from historical day  $d$ . We estimate the slope coefficients and intercepts of the AR(1) model via OLS by 15-minute period, and construct daily residual blocks such that for each historical day  $d$ , residuals are collected into  $u_d = (u_{d,1}, \dots, u_{d,96})$ . We keep the residual vectors  $u_d = (u_{d,1}, \dots, u_{d,96})$  for the rest of model estimation, which are resampled with replacement to preserve within-day correlation and spike clustering. The advantage of this semi-parametric model is to preserve the intra-day price distribution in model estimation, which parametric error distributions often fail to achieve. Overall, we are able to fit the key intra-day statistics of prices well, as shown in Appendix Figure A13.

### 5.2 Cycling cost and round-trip efficiency of batteries

Because most of the batteries operate on both the real-time energy market and the day-ahead ancillary service market, we first identify batteries that operate exclusively on the real-time energy market. Appendix Figure A14 summarizes this exercise. Among 206 batteries operating in ERCOT between March 2024 and February 2025, we identify 23 battery–season pairs that participate meaningfully and almost exclusively in the real-time market. We focus on this subset for estimation.

We calibrate each battery’s round-trip efficiency directly from the data by imposing that total net discharge to the grid equals zero over the three-month sample period. We set the annual discount factor to 0.85, which implies a per-15-minute discount factor of  $\beta = 0.85^{\frac{1}{365 \times 96}}$ .

The unit cycling cost,  $\zeta$ , is the key parameter of interest in the estimation. Together with the expected price process, it determines the optimal cycling behavior of each battery, in particular the number of charge–discharge cycles per day. Holding the cycling cost fixed, greater price volatility and a wider support of expected prices increase arbitrage opportunities, leading batteries to cycle more frequently. Conversely, for a given price process, a higher unit cycling cost reduces cycling frequency by making charge–discharge activity more costly in terms of battery degradation. Therefore, because the expected price process is estimated in the first stage, the observed number of cycles disciplines and identifies the magnitude of the cycling cost for each battery.

We estimate the unit cycling cost  $\zeta$  using a simulated method of moments (SMM), matching model-predicted battery operation to observed behavior. Simulating model-predicted battery operation requires solving dynamic programming problems according to equations (6) and (11). Because we assume the dynamic problem to be non-stationary within a day but stationary across days within each season, we solve value functions iteratively in the outer loop, and solve intra-day value functions via backward induction.

We use  $s$  to index simulated price paths. For a given  $\zeta$ , the model generates simulated quantities  $q_{dhs}(\zeta)$  (e.g., energy charged or discharged) that depend on the optimal bidding policies implied by the dynamic model. Let  $\bar{q}_h$  denote the empirical average quantity in interval  $h$ . For each simulated day  $d$  and simulation draw  $s$ , we define the discrepancy between simulated and observed quantities across all intraday periods as

$$\ell_{ds}(\zeta) = [\sum_h (q_{dhs}(\zeta) - \bar{q}_h)]^2.$$

We simulate  $D = 700$  days and  $S = 500$  independent price paths, and construct the sample objective function as the average loss:

$$Q(\zeta) = \frac{1}{DS} \sum_{d=1}^D \sum_{s=1}^S \ell_{ds}(\zeta) = \frac{1}{DS} \sum_{d=1}^D \sum_{s=1}^S [\sum_h (q_{dhs}(\zeta) - \bar{q}_h)]^2.$$

We estimate the cycling cost parameter as  $\hat{\zeta} = \arg \min_{\zeta} Q(\zeta)$ , implemented using simulated battery paths generated from the estimated price process and the model-implied bidding policies.

### 5.3 Simulation of market equilibrium

With the price process and cycling cost estimated in the first stage, we simulate the market equilibrium in which battery policies and equilibrium prices are jointly determined. Our approach operationalizes the equilibrium definition of Section 4.3: each battery bids optimally given its beliefs about the price process, the aggregate net dispatch of all batteries shifts the market-clearing condition along the non-renewable supply curve  $S_t(p_t)$ , and the resulting price process is consistent with those beliefs. We construct  $S_t(p_t)$  directly from the ERCOT bid data, and solve for the fixed point via iteration over the price-process parameters.

We simulate the 23 batteries identified in Section 6.2 as operating exclusively in the real-time energy market. Each battery is characterized by its estimated round-trip efficiency  $\delta$ , energy capacity (MWh), nameplate power capacity (MW), and estimated cycling cost  $\hat{\zeta}$ . We use system prices and aggregate battery output data from 2024 through December 4, 2025, the end of our sample as

described in Section 2.3.<sup>12</sup> For the non-renewable supply curve  $S_t(p_t)$ , we use the average aggregated non-renewable supply curves measured at noon during the summer months (June, July, and August) of 2024. We plan to use richer supply curves in our next steps.

We construct an initial sequence of equilibrium net demand by inverting the supply curve at empirical prices:  $X_t^B + X_t^{NB} = S_t(p_t^m)$ , where  $p_t^m$  denotes the observed market-clearing prices from the 2024–2025 data. We then estimate the initial price-process parameters  $\xi_h$  and  $\phi_h$  from these empirical prices; these estimates serve as the starting values for the iterative procedure.

We next recover the non-battery net demand series  $X_t^{NB}$ . Using the estimated price process, we solve the optimal bidding problem under last-minute bidding to obtain battery policies, and simulate battery net demand  $X_t^{B,\text{sim}}$ , defined as total battery load minus total battery generation at each interval  $t$  and taking empirical prices as given. We then set  $X_t^{NB} = S_t(p_t^m) - X_t^{B,\text{sim}}$ . This  $X_t^{NB}$  series is held fixed across all counterfactual scenarios, so that differences in simulated prices between the last-minute bidding and 75-minute lead window cases reflect only the change in battery bidding behavior. To prevent spurious spikes in  $X_t^{NB}$  arising from many batteries dispatching simultaneously, we apply Gaussian smoothing to  $X_t^{B,\text{sim}}$  within a 30-minute window centered on each dispatch interval before subtracting it from  $S_t(p_t^m)$ . We additionally clip  $X_t^{NB}$  to lie within the support of the supply curve.<sup>13</sup>

Given  $X_t^{NB}$  and the current price-process parameter estimates, we iterate as follows.

**Step 1.** Using the current estimates  $\xi_h$  and  $\phi_h$ , we solve the battery dynamic programming problem to obtain optimal bidding policies. Since we assume a non-stationary problem within a day and a stationary problem across days within each season, we use backward induction within each day, while iterating on the value function at the terminal period (the 96th interval) in the outer loop until convergence.

**Step 2.** We simulate market-clearing prices sequentially. At each interval  $t$ , we find the price  $p_t^{\text{new}}$  satisfying the market-clearing condition (12), i.e.,

$$S_t(p_t^{\text{new}}) = X_t^B + X_t^{NB}.$$

Batteries follow threshold-based dispatch rules. A battery at the margin of discharge is fully dispatched when its discharging cost is below that of alternative non-renewable supply. Similarly, a battery at the margin of charge is charged to full capacity when charging exceeds half of its capacity. When different batteries are simultaneously at the margins of charging and discharging, the charging battery is fully charged, while the discharging battery follows the standard dispatch

<sup>12</sup>We exclude days on which any 15-minute interval price exceeded \$1,000/MWh, since days with extreme prices are outliers that would otherwise distort the estimation of the price process parameters; this criterion removes 14 days from the sample.

<sup>13</sup>The support for the current supply curve that we are using is between 2,873 MW and 55,549 MW.

rule. The cleared price  $p_t^{\text{new}}$  then serves as the lagged price on which batteries condition their bids in interval  $t + 1$ .

**Step 3.** We re-estimate the price-process parameters  $\xi_h$  and  $\phi_h$  from the updated price series  $\{p_t^{\text{new}}\}$ .

We assess convergence using the maximum relative change across all parameters, with a floor of 0.1 applied to the denominator to handle near-zero values:

$$\max_k \frac{|\theta_k^{\text{new}} - \theta_k|}{\max(|\theta_k|, 0.1)}.$$

If this criterion exceeds 0.1, we update the price-process parameters and return to Step 1.

## 5.4 Accommodate lead time in dynamic bidding

Accommodating lead time in dynamic bidding problem is empirically challenging because batteries track all interim bid prices that are pre-committed  $\mathcal{B}_{h-z}$ , which is relatively high-dimensional. For example, if we consider a bid lead time of 75 minutes, then  $\mathcal{B}_{h-z}$  adds 10 more state variables, which imposes a heavy burden on computation.<sup>14</sup> To keep tractability of the dynamic problem, we construct an alternative set of state variables  $\Phi(s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z})$  to reduce the computation burden. The choice of  $\Phi(s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z})$  is motivated by the profit function (10). Instead of tracking  $s_{h-z}$  and  $\mathcal{B}_{h-z}$  individually, we track the probability distribution of  $s_h$ . Let the state of charge  $s_h$  lie on a discrete grid  $\mathcal{S} = \{s^1, s^2, \dots, s^N\}$ , where  $0 = s^1 < s^2 < \dots < s^N = 1$ .<sup>15</sup> We assume that batteries charge and discharge at their nameplate power capacity, implying that the number of discretized grids  $N = \frac{4}{\text{Battery duration (hours)}}$ . Rather than tracking the realized state of charge directly, we track its conditional distribution at the time when the bid is submitted. Specifically, we define

$$\tau_h = (\tau_h^1, \tau_h^2, \dots, \tau_h^N), \quad \tau_h^n = \Pr(s_h = s^n).$$

Thus,  $\tau_h$  summarizes the battery's belief about its state of charge when the realized  $s_h$  is not yet known because of the bid lead time.<sup>16</sup> Accordingly, the relevant state can be written as

<sup>14</sup>If we choose 5 grid points for each state variable, then the total state space will consist of 244 million ( $5^{12}$ ) combined grid points.

<sup>15</sup>In our baseline specification,  $N = 5$  and  $\mathcal{S} = \{0, 0.25, 0.5, 0.75, 1\}$ , but the framework applies to any discretization.

<sup>16</sup>An alternative would be to track only the conditional mean,  $\bar{s}_h = \mathbb{E}_{h-z}[s_h]$ . However, this is generally insufficient because the profit function is nonlinear in  $s_{h-z}$  and  $\mathcal{B}_{h-z}$ . Tracking only the mean therefore discards information about the distribution of future states and misses the curvature of the profit function. For this reason, we use the full

$$\Phi(s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z}) = \{\tau_h, p_{h-z-1}\}.$$

The distribution  $\tau_h$  evolves according to the battery's charging and discharging outcomes. Let  $T^g$  denote the transition matrix associated with discharge,  $T^l$  the transition matrix associated with charge, and  $I$  the identity matrix corresponding to no dispatch. Then the law of motion for beliefs can be written as

$$\tau_{h+1} = (1 - G_h^z(b_h^g))\tau_h T^g + G_h^z(b_h^l)\tau_h T^l + (G_h^z(b_h^g) - G_h^z(b_h^l))\tau_h I.$$

In the baseline grid with equally spaced increments, these transition matrices shift probability mass across adjacent state-of-charge points, with boundary adjustments at empty and full states.<sup>17</sup> The profit function and value function can then be expressed as functions of the belief state  $\tau_h$ .

$$\begin{aligned} \mathbb{E}_h[\pi(b_h^g, b_h^l, s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z})] &= \mathbb{E}_h[\pi(b_h^g, b_h^l, p_{h-z-1}, \tau_h)] \\ &= \sum_{n=1}^N [q_h^g(s_h) \times \tau_h^n] \int_{p_h \geq b_h^g} p_h dG_h^z(p_h) - \sum_{s_h} [q_h^l(s_h) \times \tau_h^n] \int_{p_h \leq b_h^l} p_h dG_h^z(p_h) \\ &\quad - \sum_{n=1}^N \zeta \times [q_h^g(s_h) \times \tau_h^n \times (1 - G_h^z(b_h^g)) + q_h^l(s_h) \times \tau_h^n \times G_h^z(b_h^l)]. \end{aligned}$$

$$V_{h-z}(\tau_h, p_{h-z-1}) = \max_{b_h^g, b_h^l} \mathbb{E}_h[\pi(b_h^g, b_h^l, \tau_h, p_{h-z-1})] + \beta \mathbb{E}_h[V_{h-z+1}(\tau_{h+1}, p_{h-z}) | b_h^g, b_h^l, \tau_h, p_{h-z-1}].$$

To solve batteries' optimal bid prices under various lead time, we first construct  $G_h^z(p_h | p_{h-z-1})$  according to the underlying one-period price transition kernels  $F_{h-k}(\cdot | \cdot)$ . To visualize the price forecast under bid lead time, we simulate future price distributions conditional on mean realized prices each period. As shown in Appendix Figure A16, the mean future prices under various lead

distribution  $\tau_h$  as the state variable.

<sup>17</sup> For example, if we discretize the space of state of charge  $s_h$  into five values,  $\{0, 0.25, 0.5, 0.75, 1\}$ , we track the following five state variables.

$$\tau_h = (\tau_h^1 = P(s_h = 0), \tau_h^2 = P(s_h = 0.25), \tau_h^3 = P(s_h = 0.5), \tau_h^4 = P(s_h = 0.75), \tau_h^5 = P(s_h = 1)).$$

The transition of  $\tau_h$  follows the evolution rules below.

$$\tau_{h+1} = (\tau_{h+1}^1, \tau_{h+1}^2, \tau_{h+1}^3, \tau_{h+1}^4, \tau_{h+1}^5) = \begin{cases} (\tau_h^1 + \tau_h^2, \tau_h^3, \tau_h^4, \tau_h^5, 0), & 1 - G_h^z(b_h^g) \\ (0, \tau_h^1, \tau_h^2, \tau_h^3 + \tau_h^4, \tau_h^5), & G_h^z(b_h^l) \\ (\tau_h^1, \tau_h^2, \tau_h^3, \tau_h^4, \tau_h^5), & G_h^z(b_h^g) - G_h^z(b_h^l) \end{cases}.$$

time follow closely with the mean future prices without lead time (Panel (a)), while the dispersion of projected prices is much larger as the lead time becomes longer (Panel (b)). Therefore, bid lead time substantially increases the likelihood of tails in the projection of future prices.

## 5.5 Estimation results and model fit

The cycling cost estimates are shown in Table 1. These estimates are jointly disciplined by the estimated price process and the observed number of daily cycles in the data. We find that most batteries in our sample have cycling costs between \$4 and \$20/MWh, broadly consistent with engineering benchmarks such as He et al. (2020). One notable exception is Russek Street BESS, for which we estimate a substantially higher cycling cost. This reflects the combination of persistently high electricity prices at its settlement point and a relatively low observed cycling frequency, which together imply that the battery forgoes profitable arbitrage opportunities and is consistent with a high marginal cycling cost.

More broadly, as shown in Figure 5, we find that cycling costs are higher for batteries with lower average daily discharge relative to capacity. Batteries with longer durations tend to exhibit slightly lower cycling costs, while we find little to no correlation between cycling cost and nameplate power capacity (MW).

We show our model fit by simulating operations of sampled batteries and compare with the observed data. We pick one battery, Casl Gap, a one-hour battery with nameplate capacity of 10 MW.<sup>18</sup> We calibrate its round-trip efficiency as 0.88 to make total charging and discharging balanced. The comparison between simulated charging/discharging pattern and data is shown in Appendix Figure A15. Overall, our model captures the daily charging and discharging patterns well.

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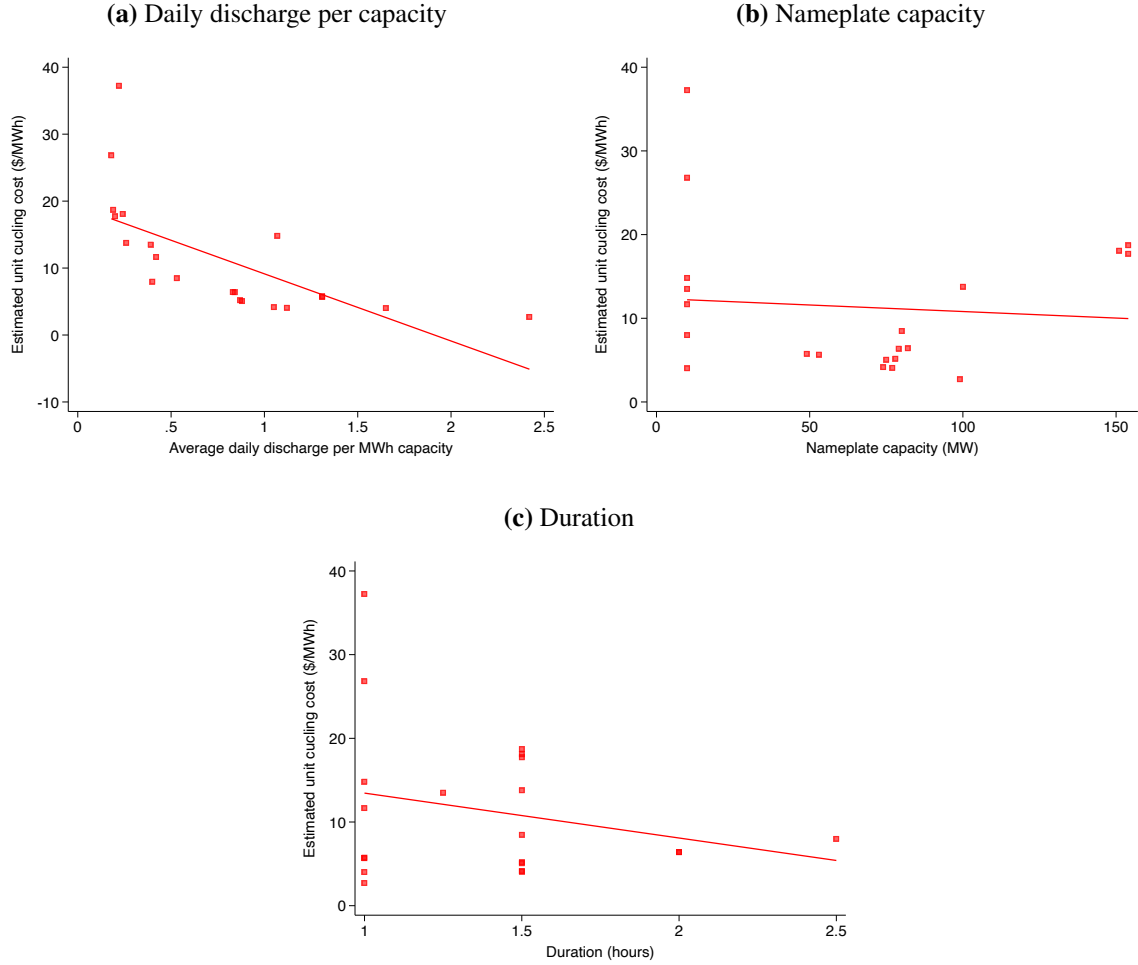
<sup>18</sup>Officially, it's reported as a 4.2-hour battery. However, from its charging and discharging pattern, it operates like a one-hour battery.

**Table 1: Estimated Cycling Costs for ERCOT Battery Storage Units**

| Resource          | Sample Period     | Capacity (MW) | Duration (h) | Cost (\$/MWh) | SE   |
|-------------------|-------------------|---------------|--------------|---------------|------|
| BIG_STAR_BESS     | Dec 2024–Feb 2025 | 80            | 1.5          | 8.48          | 0.02 |
| BR_ARROW_BESS1    | Dec 2024–Feb 2025 | 49            | 1            | 5.76          | 0.01 |
| BR_ARROW_BESS2    | Dec 2024–Feb 2025 | 53            | 1            | 5.67          | 0.01 |
| CASL_GAP_BATTERY1 | Dec 2024–Feb 2025 | 10            | 1            | 14.83         | 0.08 |
| CR_BESS1          | Dec 2024–Feb 2025 | 10            | 2.5          | 7.99          | 0.02 |
| DAN_BESS1         | Dec 2024–Feb 2025 | 78            | 1.5          | 5.18          | 0.02 |
| DAN_BESS2         | Dec 2024–Feb 2025 | 75            | 1.5          | 5.06          | 0.01 |
| ELZA_SLR_BES1     | Dec 2024–Feb 2025 | 100           | 1.5          | 13.78         | 0.05 |
| GREGORY_BESS1     | Dec 2024–Feb 2025 | 10            | 1            | 11.66         | 0.03 |
| INRT_W_BESS_1     | Dec 2024–Feb 2025 | 13            | 3.5          | 7.03          | 0.02 |
| JADE_SLR_BESS1    | Dec 2024–Feb 2025 | 79            | 2            | 6.38          | 0.02 |
| JADE_SLR_BESS2    | Dec 2024–Feb 2025 | 82            | 2            | 6.41          | 0.02 |
| JAR_BES1          | Dec 2024–Feb 2025 | 154           | 1.5          | 17.72         | 0.09 |
| JAR_BES2          | Dec 2024–Feb 2025 | 154           | 1.5          | 18.72         | 0.09 |
| LIGSW_BESS1       | Dec 2024–Feb 2025 | 10            | 1.25         | 13.52         | 0.03 |
| MIDWY_BESS1       | Jun 2024–Aug 2024 | 10            | 1            | 37.25         | 0.09 |
| MUENSTER_BESS1    | Dec 2024–Feb 2025 | 10            | 1            | 26.82         | 0.09 |
| MYR_BES1          | Dec 2024–Feb 2025 | 77            | 1.5          | 4.08          | 0.01 |
| MYR_BES2          | Dec 2024–Feb 2025 | 74            | 1.5          | 4.16          | 0.01 |
| PHO_BES1          | Dec 2024–Feb 2025 | 151           | 1.5          | 18.09         | 0.08 |
| RUSSEKST_BESS     | Dec 2024–Feb 2025 | 10            | 1            | 94.36         | 0.36 |
| SHAMROCK_BESS1    | Dec 2024–Feb 2025 | 99            | 1            | 2.71          | 0.01 |
| TV_BESS           | Dec 2024–Feb 2025 | 10            | 1            | 4.02          | 0.01 |

Notes: This table shows the estimated unit cycling cost of 23 battery–season pairs that participate meaningfully and almost exclusively in the real-time market.

**Figure 5: Cycling costs across batteries**



Notes: This figures shows the estimated unit cycling cost of 23 battery–season pairs that participate meaningfully and almost exclusively in the real-time market, as well as its correlation with average daily discharge per energy capacity (in MWh) in Panel (a), nameplate capacity (in MW) in Panel (b), and battery durations (in hours) in Panel (c). We drop Russek Street BESS since its estimated unit cycling cost is an outlier, and we drop one battery that has a duration above 3 hours.

## 6 Counterfactual Simulations

We use the estimated structural model to answer three key questions. First, focusing on one representative battery, we examine how bid lead times shape bidding behavior and profitability, and how these effects interact with technology characteristics and price volatility. Second, we evaluate a policy alternative that allows batteries to submit state-of-charge–dependent bids, and assess its effectiveness in mitigating efficiency losses. Third, we embed these policy counterfactuals in a

market equilibrium framework to quantify how bidding inefficiencies affect grid outcomes and, more broadly, the energy transition.

## 6.1 The impacts of bid lead time

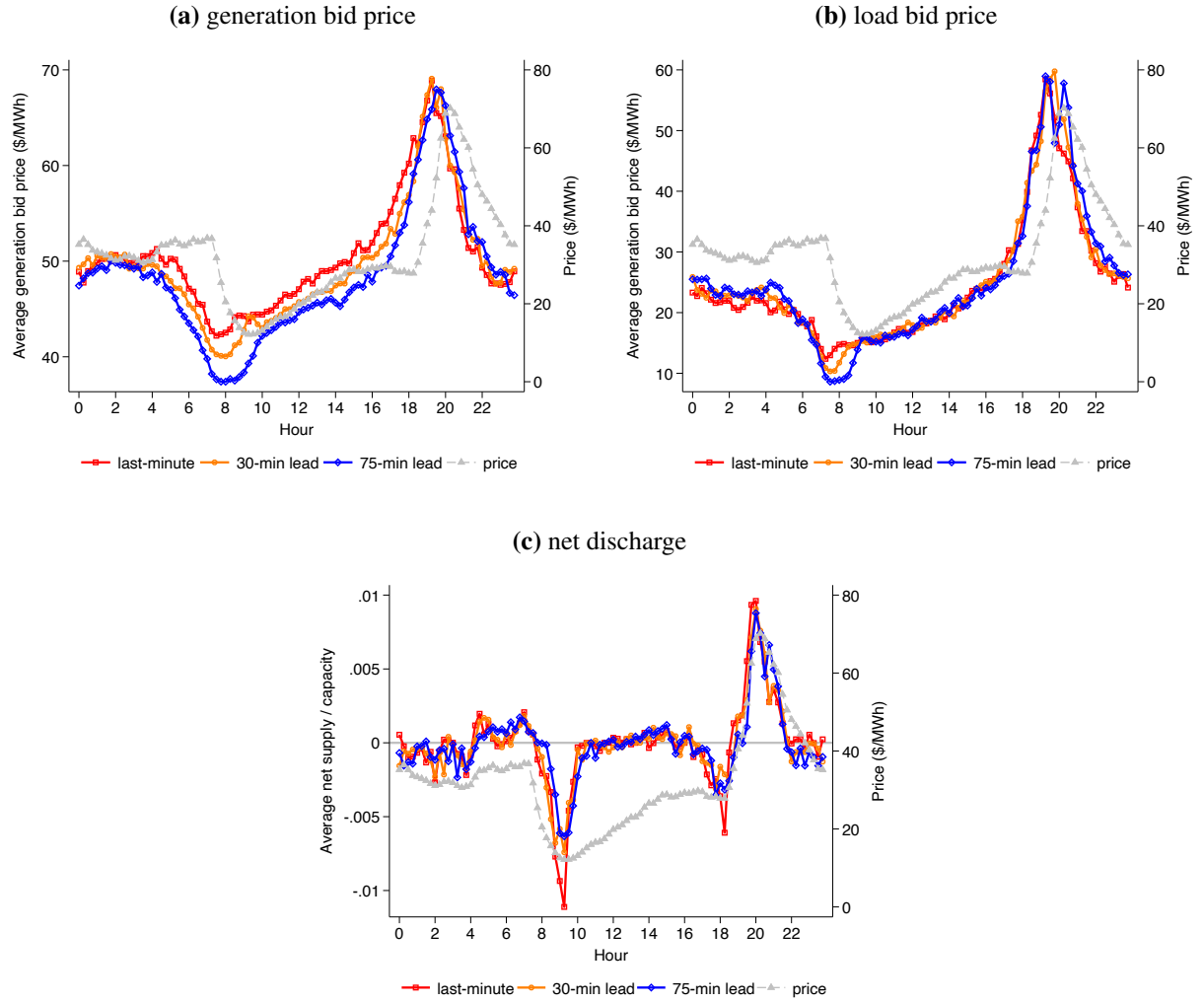
We first investigate how lead time affects batteries' bidding strategies, net energy supply, and profits. We solve the generation and load bid prices under various bid lead time using parameters of Casl Gap, and then simulate operations of this one-hour batteries with 10 MW power capacity for 300 days by 1,000 times. The results are shown in Figure 6. Relative to last-minute bidding, batteries tend to submit lower generation bid prices throughout the day under longer lead time. This occurs because a larger dispersion in the projected price distribution increases the density of tails, which are the primary drivers of batteries' expected profits. Consequently, batteries are more likely to be discharged earlier in the day but supply less energy during peak-price periods, leading to a substantial deviation from optimal charging-discharging pattern under last-minute bidding.

Moreover, we document a sharp profit-lead horizon gradient as shown in Figure 7. The baseline simulated profit under zero lead time is \$10.4 per kW of installed capacity, which is close to the \$18.7 energy arbitrage revenue per kW of installed capacity as calculated by Modo Energy. Under 30-minute bid lead time as currently implemented in MISO, SPP, and ISO-NE, the average profit is reduced to \$8.6 per kW, which is only 82.6% of the baseline profit. Under 75-minute bid lead time as currently implemented in CAISO, the average profit is reduced to \$7.9 per kW, which is only 75.7% of the baseline profit. These results highlights the importance of bidding horizon in shaping operational profits for batteries.

We further examine how the profit impacts of bid lead time vary with technology characteristics and price volatility. As shown in Figure 8, longer-duration batteries partially mitigate the distortions induced by bid lead time. The reduction in energy arbitrage profits declines from 24.3% for one-hour batteries to 11.7% for four-hour batteries, reflecting the greater inter-temporal flexibility of longer-duration storage. Given that more than 80% of installed capacity in CAISO consists of four-hour batteries, the adverse effects of bid lead time on profits are attenuated in practice, though they remain economically meaningful.

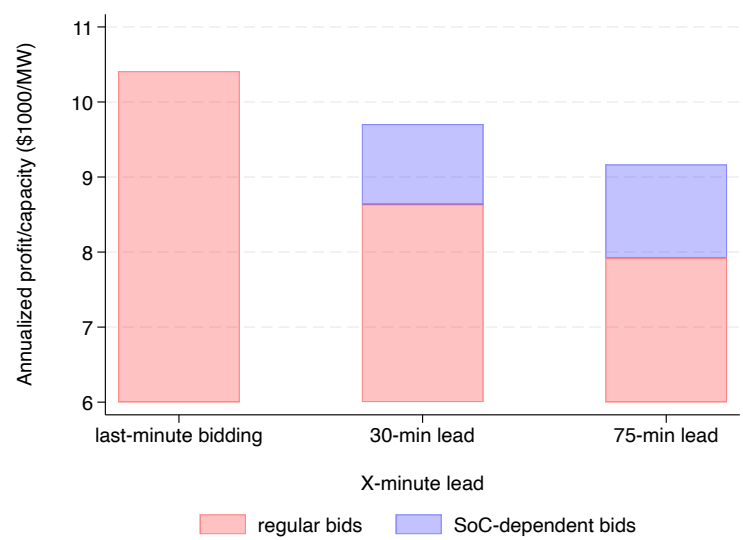
In contrast, the impact of bid lead time is amplified by greater price volatility. As shown in Panel (b), a 10% increase in price volatility raises profit losses from 24.3% to 28.8%. These results suggest that bid lead time becomes a more consequential market design friction as volatility increases, particularly in systems with higher renewable penetration where price fluctuations are more pronounced.

**Figure 6: Bids, Energy Supply, Profits under Bid Lead Time**



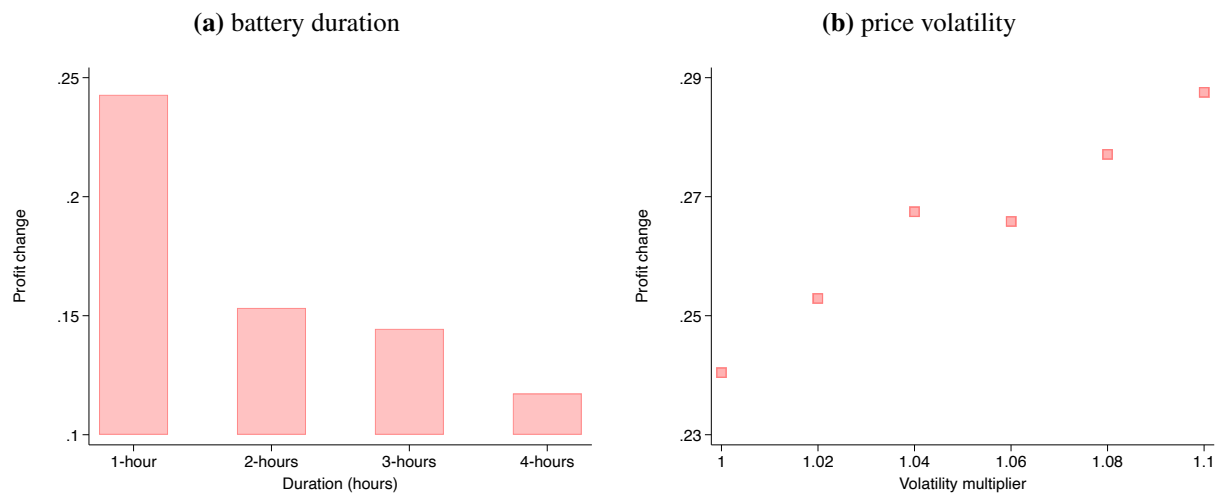
Notes: This figure plots the simulated generation bid prices, load bid prices, net energy supply, and profit at the 15-minute frequency for three scenarios: (1) last-minute bidding (red squares), (2) 30-minute bid lead time (orange circles), and (3) 75-minute bid lead time (blue diamonds). We further plot the real-time market prices in gray.

**Figure 7:** Profits under Bid Lead Time and SoC-dependent Bids



Notes: This figure plots the simulated profits under various bid lead time conditional on mean realized prices each period.

**Figure 8:** Profit Changes under Bid Lead Time: the Role of Durations and Volatility



Notes: This figure plots the profit impacts of bid lead time under different battery durations (Panel (a)) and price volatility (Panel (b)). The horizontal axis of Panel (b) denotes the multipliers we apply to the residuals estimated from equation (3).

## 6.2 State-of-charge-dependent bids

We introduce state-of-charge-dependent bids as a policy remedy discussed by the CAISO Market Surveillance Committee as we highlight in Section 2. Instead of submitting  $b_h^g$  and  $b_h^l$ , batteries can submit bids as a function of  $s_h$ , denoted by  $b_h^g(s_h)$  and  $b_h^l(s_h)$ . Compared with 10, the new profit function is as follows and the key difference is that now the bid price is SoC dependent.

$$\begin{aligned} & \mathbb{E}_h[\pi(b_h^g(s_h), b_h^l(s_h), s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z})] \\ &= \sum_{s_h} [q_h^g(s_h) \times \tau_h^{s_h}] \times \int_{p_h \geq b_h^g(s_h)} p_h dG_h^z(p_h) - \sum_{s_h} [q_h^l(s_h) \times \tau_h^{s_h}] \times \int_{p_h \leq b_h^l(s_h)} p_h dG_h^z(p_h) \\ & \quad - \sum_{s_h} \frac{\zeta \times [q_h^g(s_h) \times \tau_h^{s_h} \times (1 - G_h^z(b_h^g(s_h))) + q_h^l(s_h) \times \tau_h^{s_h} \times G_h^z(b_h^l(s_h))]}{k}. \end{aligned}$$

However, allowing full flexibility of  $b_h^g(s_h)$  and  $b_h^l(s_h)$  makes the transition process of  $\tau_h$  very challenging to specify and track. Therefore, we impose more structures on the state-of-charge dependent bids as follows.

$$b_h^g = \begin{cases} b_h^{g1}, & s_h < 0.5 \\ b_h^{g2}, & s_h \geq 0.5 \end{cases} \quad b_h^l = \begin{cases} b_h^{l1}, & s_h > 0.5 \\ b_h^{l2}, & s_h \leq 0.5 \end{cases}$$

We categorize possible state-of-charge into two types: low state of charge when state of charge is below 0.5, and high state of charge when state of charge is above 0.5. Moreover, we impose another assumption such that  $b_h^{g1} \geq b_h^{g2} \geq b_h^{l2} \geq b_h^{l1}$ . This monotonicity assumption helps us simplify the expression of state of charge transition dynamics. Moreover, this is intuitive since when state of charge is low, batteries are more conservative in bidding to discharge, and when state of charge is high, batteries are more conservative in bidding to charge. Under this assumption, we can write the transition dynamics of as follows.

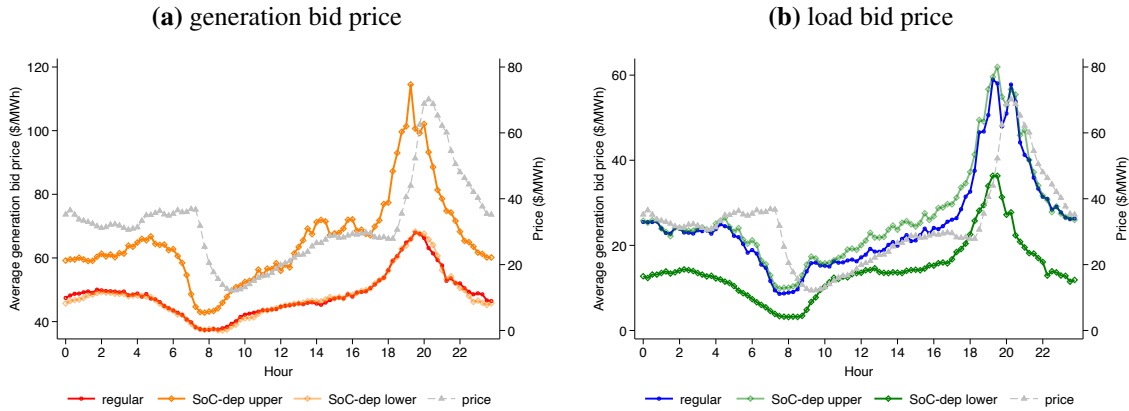
$$\tau_{h+1} = \begin{cases} (\tau_h^1 + \tau_h^2, \tau_h^3, \dots, \tau_h^N, 0), & 1 - G_h^z(b_h^{g1}), \\ (\tau_h^1, \dots, \tau_h^{m-1}, \tau_h^m + \tau_h^{m+1}, \tau_h^{m+2}, \dots, \tau_h^N, 0), & G_h^z(b_h^{g1}) - G_h^z(b_h^{g2}), \\ (\tau_h^1, \tau_h^2, \dots, \tau_h^N), & G(b_h^{g2}) - G_h^z(b_h^{l2}), \\ (0, \tau_h^1, \dots, \tau_h^{m-1}, \tau_h^m + \tau_h^{m+1}, \tau_h^{m+2}, \dots, \tau_h^N), & G(b_h^{l2}) - G_h^z(b_h^{l1}), \\ (0, \tau_h^1, \tau_h^2, \dots, \tau_h^{N-2}, \tau_h^{N-1} + \tau_h^N), & G_h^z(b_h^{l1}). \end{cases}$$

The value function can be written down as follows.

$$V_{h-z}(\tau_h, p_{h-z-1}) = \max_{b_h^g, b_h^l} \mathbb{E}_h[\pi_h(b_h^g(s_h), b_h^l(s_h), \tau_h, p_{h-z-1})] \\ + \beta \mathbb{E}_h[V_{h-z+1}(\tau_{h+1}, p_{h-z}) | b_h^g(s_h), b_h^l(s_h), \tau_h, p_{h-z-1}]$$

We solve the generation and load bid prices using state-of-charge-dependent bids and under various bid lead time, and then simulate operations of the same identical one-hour batteries with 10 MW power capacity for 300 days by 1,000 times. The average generation bid prices, as shown in Panel (a) of Figure 9, differentiate substantially between the upper level ( $b_h^{g1}$ ) and lower level ( $b_h^{g2}$ ) with the regular generation bid prices closer to the lower level. This is intuitive since the potential opportunity cost of discharging will be much higher if the batteries have already been on low state of charge, and allowing state-of-charge dependent bids allow batteries to make contingent plans on realized state of charge and submit more sophisticated bids. Similarly, the average load bid prices are closer to the upper level ( $b_h^{l2}$ ) with substantial level differences between the upper level ( $b_h^{l2}$ ) and lower level ( $b_h^{l1}$ ).

**Figure 9: Bids Prices Using SoC-dependent Bids**



Notes: This figure plots the simulated generation bid prices and load bid prices under 30-minute bid lead time with regular bidding and SoC-dependent bids.

State-of-charge-dependent bids brings substantial improvement to batteries' operation profits, since they mitigate quantity uncertainty and allow contingent plans based on different realizations of state of charge. As shown in Figure 7, under 30-minute bid lead time, state-of-charge-dependent bids improve the average profit by 12.3%, and close the gap from the baseline profit by 10.2 percentage points (83.0% to 93.2%). Under 75-minute bid lead time, state-of-charge-dependent

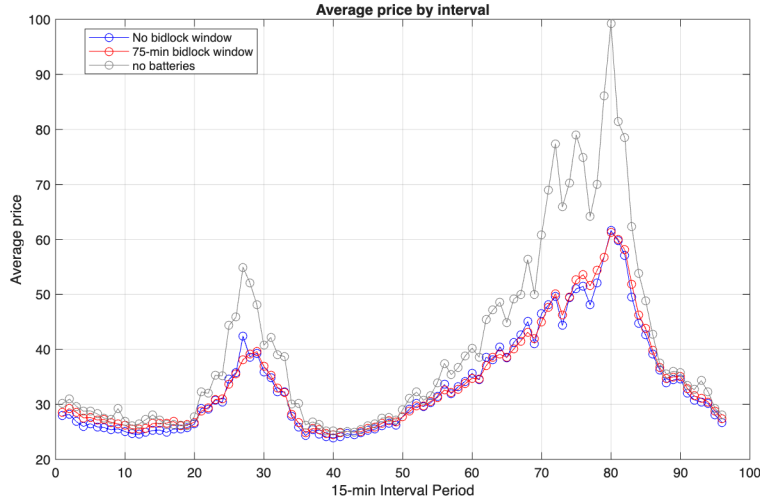
bids improve the average profit by 15.8%, and close the gap from the baseline profit by 12.0 percentage points (76.1% to 88.1%). The profit-lead horizon gradient is much flatter under state-of-charge-dependent bids. Therefore, allowing contingent bidding plans significantly lowers the uncertainty towards quantities, and improves batteries' operation profits.

### 6.3 Equilibrium consequences

We simulate the two bidding regimes, last-minute bidding and 75-minute bid lead time, in the market equilibrium framework of Section 5.3 to assess how bid lead time shapes electricity prices and aggregate welfare. We also report a no-battery counterfactual, in which battery net demand is held at zero and non-renewable supply alone clears the market, as a baseline for the overall effect of battery participation. Please notice that the results reported in this section is very preliminary and based only on 23 batteries we've estimated the cycling cost.

**Price effects.** Moving from a 75-minute bid-lock window to no bid-lock lowers equilibrium prices, as more precise bidding allows batteries to more effectively shave peak demand. As shown in Table 2 and Figure 10, the mean equilibrium price falls from \$34.2/MWh under the 75-minute window to \$33.8/MWh under no bid-lock — a reduction of about \$0.44/MWh. The effect is more pronounced in the upper tail: the 99th-percentile price falls from \$232.9 to \$224.5, a difference of \$8.4/MWh, indicating that eliminating bid lead time most acutely improves batteries' ability to respond to extreme price events. Median prices are nearly identical across the two regimes (\$22.5 vs. \$22.6/MWh), confirming that the benefit of removing the bidding constraint operates primarily through the tails of the distribution. For reference, both battery equilibria substantially reduce prices relative to the no-battery counterfactual (\$40.1/MWh mean), indicating that battery participation itself generates large benefits regardless of the bidding regime.

**Figure 10: Average Price by Interval under Bidding Counterfactuals**



Notes: This figure plots the average market-clearing price by 15-minute interval (1–96) for three scenarios: no bid-lock window (blue), 75-minute bid-lock window (red), and no batteries (gray). Prices are averaged over all days in the 2024–2025 sample.

**Table 2: Summary Statistics of Simulated Market Prices**

| Statistic       | Last-minute bidding | 75-minute lead time | No batteries |
|-----------------|---------------------|---------------------|--------------|
| 5th percentile  | 4.8                 | 4.9                 | 4.8          |
| Median          | 22.5                | 22.5                | 22.5         |
| Mean            | 33.8                | 34.2                | 40.1         |
| 95th percentile | 85.2                | 85.0                | 91.7         |
| 99th percentile | 224.5               | 232.8               | 309.6        |

Notes: This table shows the key statistics of simulate market prices under three scenarios: no bid-lock window, 75-minute bid-lock window, and no batteries.

**Welfare decomposition.** Table 3 reports average daily welfare components across the three scenarios. Removing the bid-lock window raises battery profits by roughly 9%, from \$27.8/day per MW of installed capacity under the 75-minute window to \$30.4/day, consistent with the individual-level profit gains documented in Section 6.1. Consumer surplus, measured here as the reduction

in consumers' total energy expenditure (price times non-battery net demand) relative to the no-battery counterfactual, rises by roughly \$530,000 per day, or about 7%, when the bid-lock window is removed.

The lower prices under no bid-lock compress profits for other market participants. Average daily fossil profit falls from \$32.9 million under the 75-minute window to \$32.4 million under last-minute bidding, a reduction of about \$535,000/day, with per-MWh margins falling from \$28.0 to \$27.6. Wind follows the same pattern: we measure wind and solar revenues per MW of installed capacity as the capacity factor times the simulated market-clearing price, accumulated over each 15-minute interval, where capacity factors are derived from atmospheric conditions in the ERA5 reanalysis dataset. Daily wind revenue per MW of installed capacity falls from \$344.2 (75-minute window) to \$338.4 (no bid-lock), reflecting the same broad price-lowering effect that reduces fossil margins.

However, revenue trends for solar are different; solar profits per MW are essentially flat across the two regimes (\$185.8 under no bid-lock versus \$184.4 under the 75-minute window), if not slightly more favorable in the no bid-lock scenario. The reason why is that the temporal arbitrage that batteries perform within a day, charging during midday hours when prices are low and discharging during peak evening hours, ends to be particularly complementary with solar generation, which is high during the day and offline in the evening. Under no bid-lock, batteries are more aggressive about cycling as they are more certain that they can time their charge-discharge cycles precisely around price trends. Thus, battery-solar complementarity is increased when batteries can bid more effectively under no bid-lock, roughly offsetting the price-lowering effect of more active battery arbitrage overall.

**Emissions and total surplus.** We compute emissions by constructing an emissions supply curve that pairs each fossil plant's position in the merit order with its CO<sub>2</sub> emission rate, yielding total CO<sub>2</sub> output (in lbs/MWh) as a function of aggregate non-renewable dispatch. At each simulated interval, we map the non-battery generation level onto this curve to recover total emissions, then convert to short tons. The difference in daily emissions between the two bidding regimes is negligible: 309,895 short tons under the 75-minute window versus 309,924 under no bid-lock, a gap of less than 30 tons. Applying a social cost of carbon of \$185 per short ton, the emissions channel contributes only about \$5,400/day to the welfare comparison between regimes. Summing consumer surplus, battery profits, and fossil profits and netting out the social cost of emissions, the change in daily total surplus relative to the no-battery counterfactual is \$78,615 under the 75-minute window and \$76,447 under no bid-lock. The slightly lower surplus change under no bid-lock reflects the partially offsetting effects of higher consumer surplus and lower fossil margins. Taken together, these results indicate that the primary market-level consequence of removing

bid lead time is a transfer from fossil producers to batteries and consumers, amounting to roughly \$535,000/day in consumer surplus alone.

**Table 3: Summary Statistics of Simulated Market Prices**

|                                                                    | Last-minute bidding | 75-minute lead time | No batteries |
|--------------------------------------------------------------------|---------------------|---------------------|--------------|
| Average daily battery profit per MW capacity                       | 30.4                | 27.8                | -            |
| Average daily fossil generation                                    | 1,019,569           | 1,019,512           | 1,019,310    |
| Average daily fossil generation cost                               | 6,960,252           | 6,966,204           | 7,076,014    |
| Average daily fossil profit                                        | 32,352,522          | 32,885,494          | 40,546,839   |
| Average daily fossil profit per MWh generated                      | 27.6                | 28.0                | 33.8         |
| Average daily CS (change in energy cost from no-battery)           | 8,256,657           | 7,724,114           | -            |
| Average daily wind profit per MW capacity                          | 338.4               | 344.2               | 408.4        |
| Average daily solar profit per MW capacity                         | 185.8               | 184.4               | 209.7        |
| Average daily emissions (short ton)                                | 309,924             | 309,895             | 309,771      |
| Average daily social cost from emissions                           | 57,336,023          | 57,330,592          | 57,307,642   |
| Average change in surplus from no-battery (CS + all profits - SCC) | 76,447              | 78,615              | -            |

Notes: This table shows the key statistics of the market under three scenarios: no bid-lock window, 75-minute bid-lock window, and no batteries.

## 7 Conclusion

Despite growing economic interests in energy storages, limited attention has been paid to how batteries operate in electricity markets, how efficiently they bid as energy arbitrageurs, and how bidding (in)efficiencies affect market equilibrium and the broader energy transition. This paper investigates a key market design feature of real-time electricity auctions that is particularly relevant to battery bidding: the bid lead time, which is a common design in the wholesale electricity market

around the world and varies across markets. We compile a comprehensive data set covering all utility-scale batteries in ERCOT from 2018 to 2025, and we access the real-time bidding and settlement information for every 15 minutes. Using these high-frequency data, we document three key data patterns. First, we find that batteries are actively bidding into the real-time energy market, charging during low-price periods and discharging during high-price periods to maximize profits. Batteries submit high generation bid prices to avoid being dispatched too early and to preserve the high option value of stored energy. Second, we find that batteries respond to unexpected market shocks by quickly adjusting the generation bid prices. Third, since accurate prediction of future prices is a key prerequisite for batteries to earn profits, we demonstrate a critical role of recent price information in reducing forecast errors. Motivated by these empirical facts, we develop an equilibrium model of electricity markets, which embeds a dynamic model of battery bidding. We then introduce bid lead time as a market design constraint such that bids for a given interval must be submitted several periods in advance. Overall, we find that bid timing is a quantitatively important dimension of market design: even relatively short lead times generate meaningful inefficiencies in storage operations and distort equilibrium outcomes in electricity markets.

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# **Online Appendix for Battery Bidding and Market Efficiency in Energy Transition**

Hunt Allcott      Luming Chen      Julia Park

## **Table of Contents**

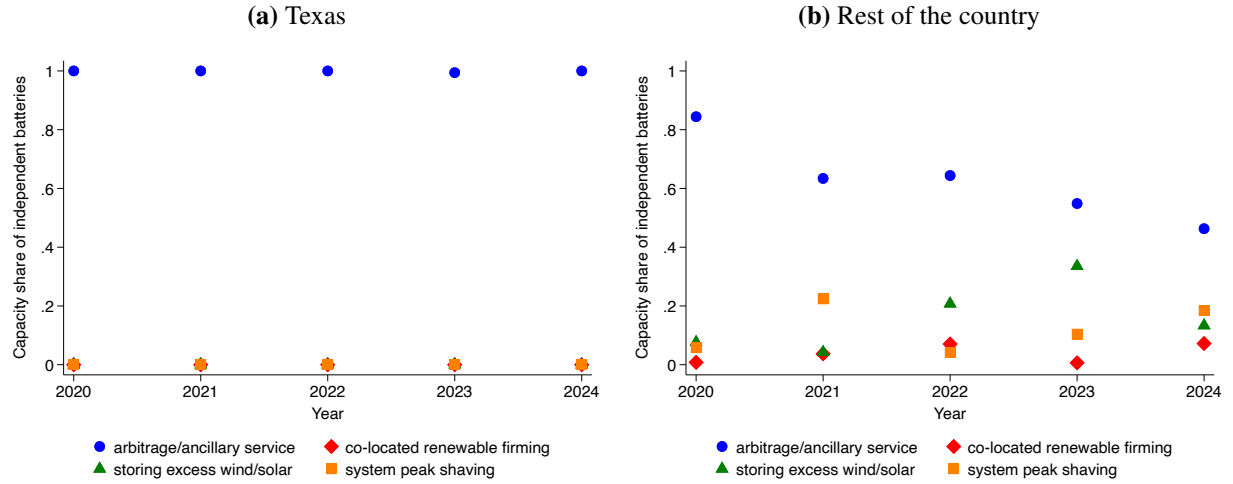
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|          |                                      |           |
|----------|--------------------------------------|-----------|
| <b>A</b> | <b>Additional Figures and Tables</b> | <b>2</b>  |
| <b>B</b> | <b>Automated bidding adoption</b>    | <b>9</b>  |
| <b>C</b> | <b>Details on model estimation</b>   | <b>12</b> |
| <b>D</b> | <b>Details on model simulation</b>   | <b>15</b> |

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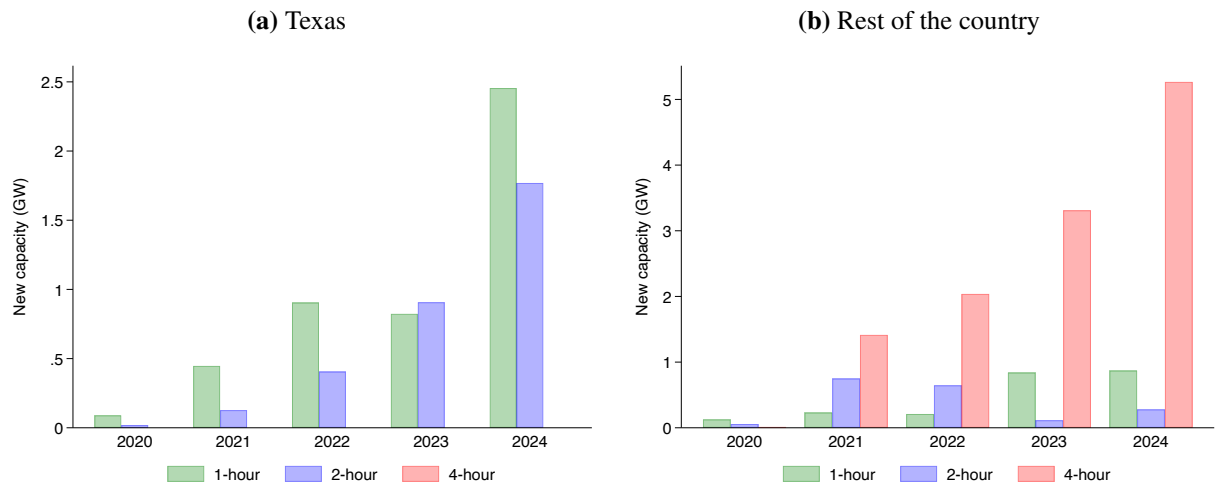
## A Additional Figures and Tables

**Figure A1: Capacity Shares of Independent Batteries**



Notes: This figure plots the capacity shares of batteries that are independent (Panel (a)) and that are not supporting other generation units (Panel (b)) from 2018 to 2023. A battery is independent, defined as not being coupled with another generators. A battery directly supports another unit (mostly solar) if it is intended for dedicated generator firming or storing excess generation of other units. We plot the capacity shares nationwide (blue circles), in California (red diamonds), and in Texas (orange squares, the focus of this paper). The data source is EIA-860.

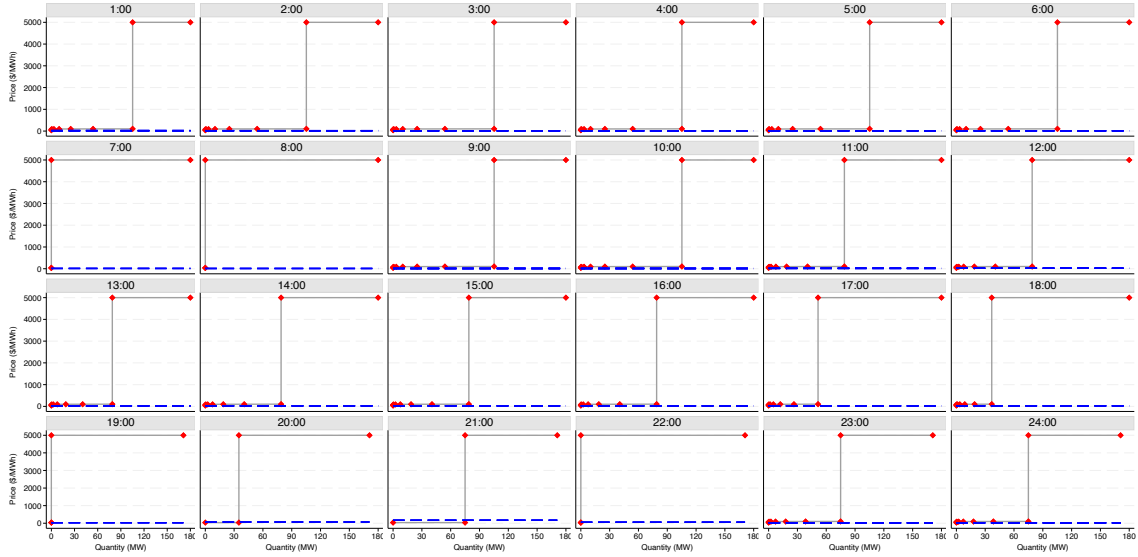
**Figure A2: New Capacity of Batteries by Durations**



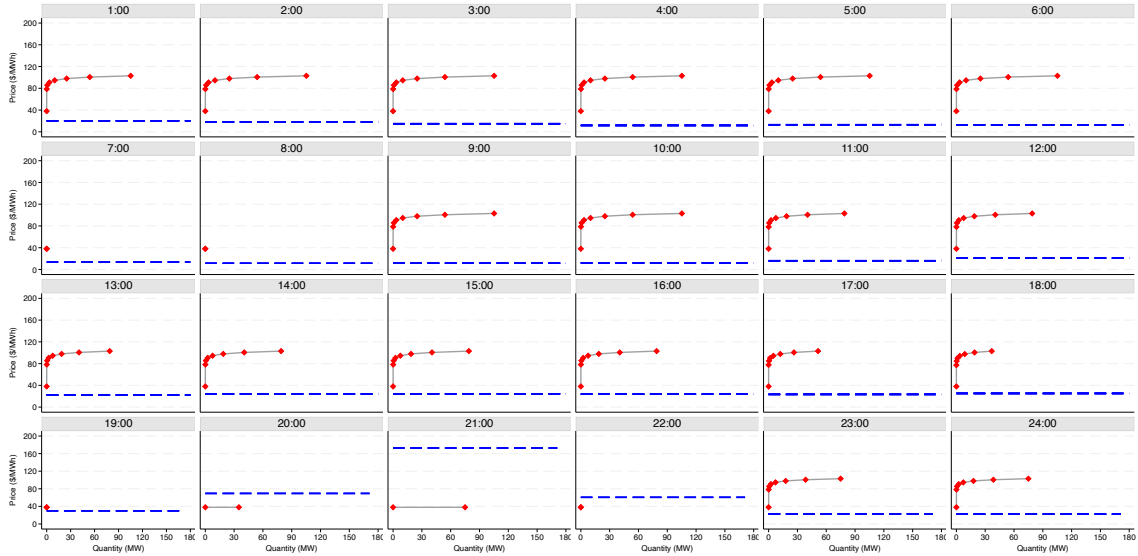
Notes: This figure plots the new capacity of batteries by durations (1-hour batteries, 2-hour batteries, 4-hour batteries) nationwide (Panel (a)) and in Texas (Panel (b)) from 2018 to 2023. A 1-hour battery could maintain its full power output for 1 hour. The data source is EIA-860.

**Figure A3: Generation Bids of Turquoise Storage on August 18, 2024**

**(a) raw pattern**



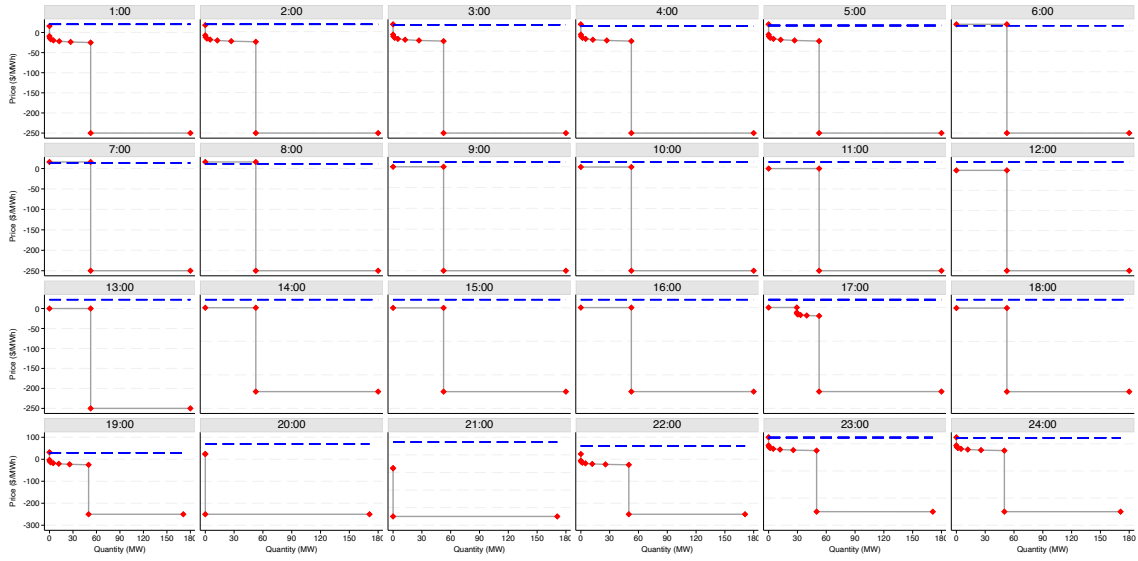
**(b) censored bid price**



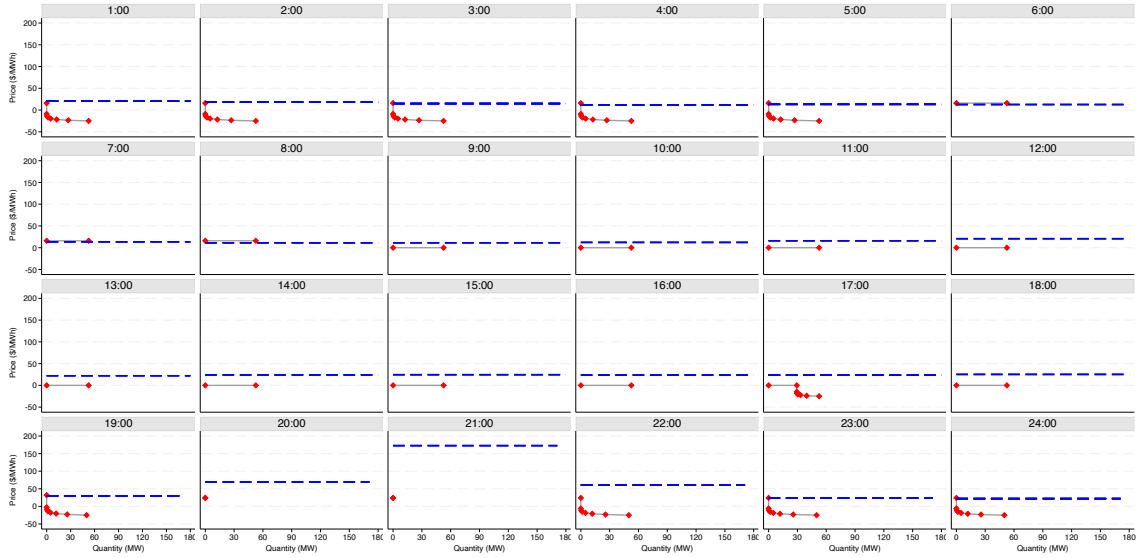
Notes: This figure plots the generation bids submitted by Turquoise Storage, one of the largest utility-scale battery in ERCOT. The bids are submitted on August 18, 2024 when Texas experiences significant heat, with temperatures reaching as high as 107°F in several locations. Panel (a) plots the raw bids and panel (b) plots the truncated bids at -10 dollars/MWh and 2,000 dollars/MWh. We use blue dashed line to denote the average real-time market prices for the corresponding hour in Texas on that day.

**Figure A4: Load Bids of Turquoise Storage on August 18, 2024**

**(a) raw pattern**

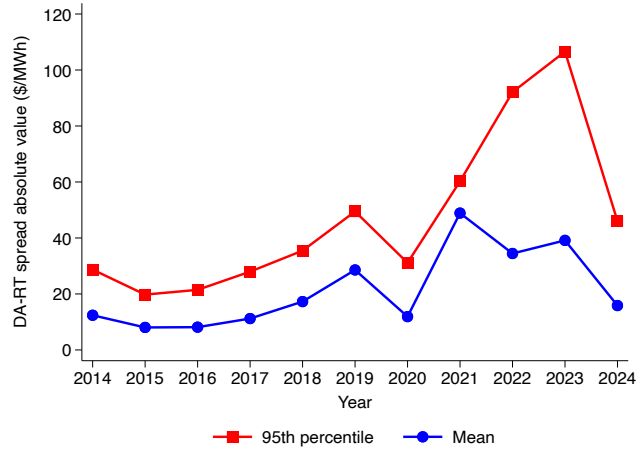


**(b) censored bid price**



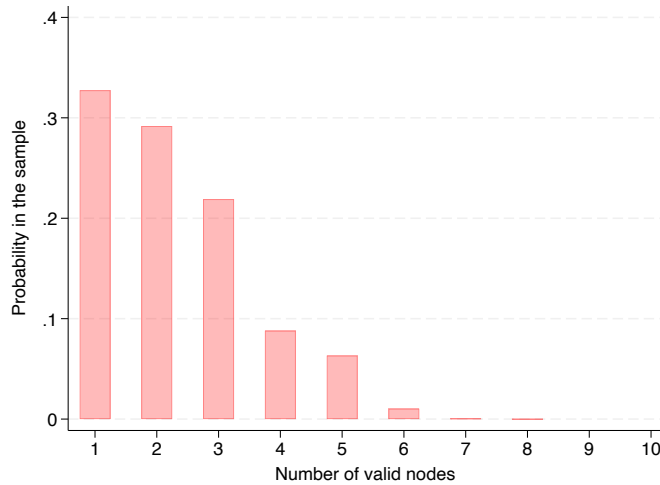
Notes: This figure plots the load bids submitted by Turquoise Storage, one of the largest utility-scale battery in ERCOT. The bids are submitted on August 18, 2024 when Texas experiences significant heat, with temperatures reaching as high as 107°F in several locations. Panel (a) plots the raw bids and panel (b) plots the truncated bids at -100 dollars/MWh and 1,000 dollars/MWh. We use blue dashed line to denote the average real-time market prices for the corresponding hour in Texas on that day.

**Figure A5: DART Spread in ERCOT**



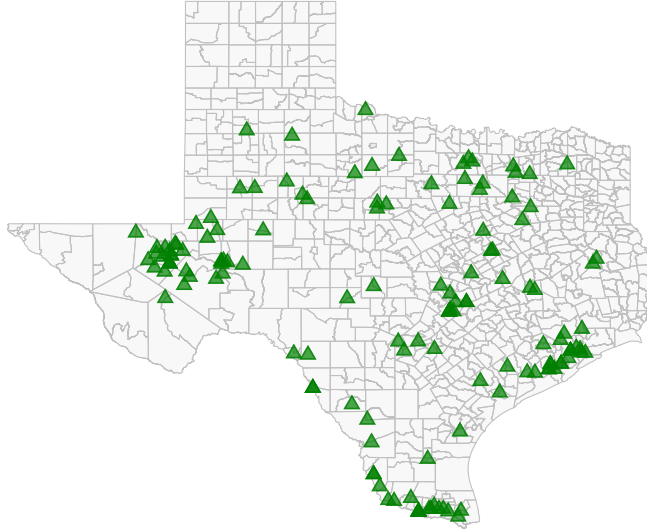
Notes: This figure plots the time series of absolute values of DART spread, which is defined as the difference between the day-ahead market price and real-time market price. We calculate the absolute values of DART spread at 15-minute frequency and calculate the annual averages (red circles) and annual 95<sup>th</sup> percentiles (blue triangles).

**Figure A7: Numbers of Valid Bidding Segments**



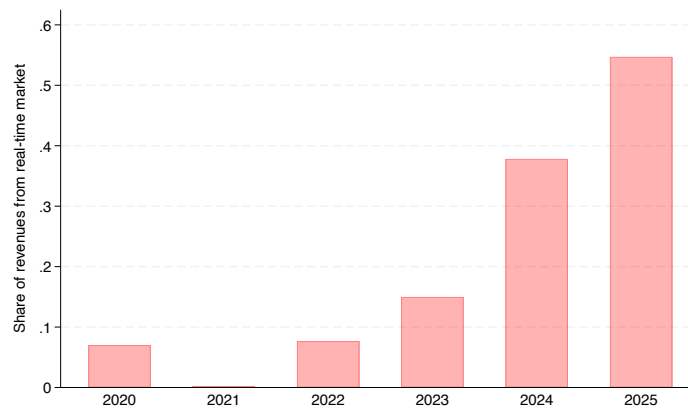
Notes: This figure plots the distribution of the numbers of valid bidding segments using bids information for each battery at 15-minute frequency in 2024. We define a bid segment as *valid* if its price lies between -\$50 and \$1,000/MWh, as the probability of prices falling outside this range is approximately 0.1% in either tail. In addition, we merge adjacent segments whose bid quantities differ by less than 0.1 MW or whose bid prices differ by less than \$0.1/MWh.

**Figure A6: Map of Batteries in ERCOT**



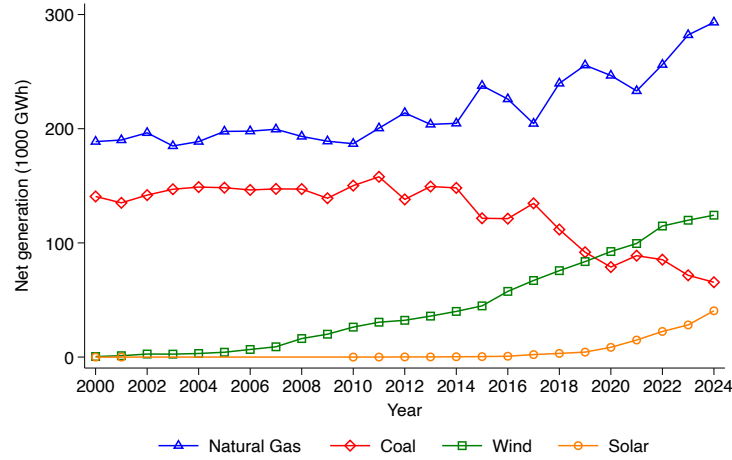
Notes: This figure plots the locations of utility-scale batteries in ERCOT in 2023. The coordinates of batteries are obtained from EIA-860.

**Figure A8: Revenue Shares from Real-time Market**



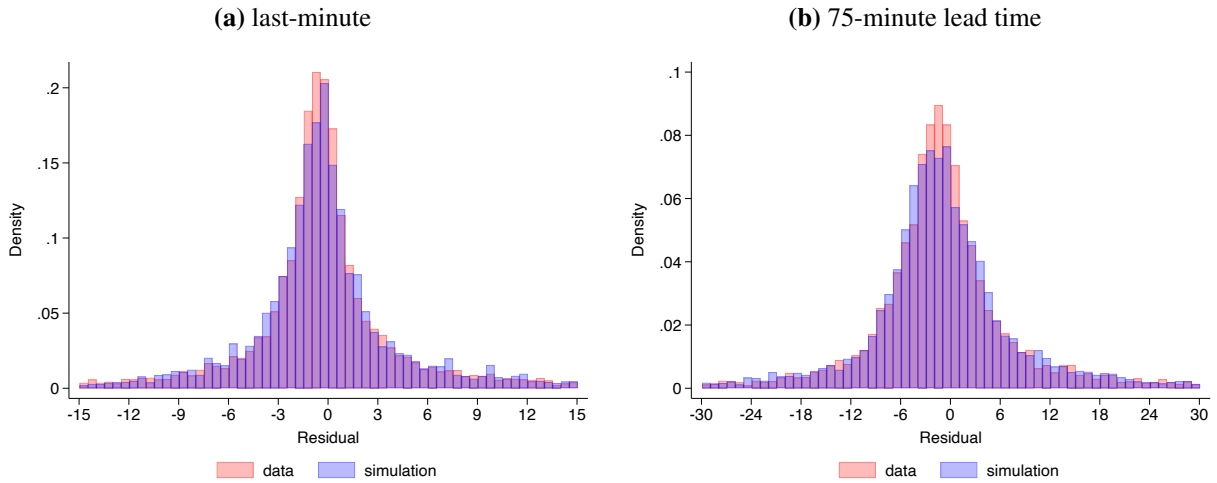
Notes: This figure plots the revenue shares from real-time market for batteries in ERCOT in 2020-2025. The revenue shares from real-time market are defined as the ratio of revenues from real-time energy arbitrage in total revenues from both real-time energy arbitrage and day-ahead ancillary service market.

**Figure A9: Energy Composition in ERCOT**



Notes: This figure plots the total electricity generation from different sources in ERCOT, compiled from EIA-923.

**Figure A10: Price Forecast Errors: Data versus Model**



Notes: This figure compares the forecast errors in price AR(1) models (equation (3)) between data and simulated prices. Panel (a) plots the distribution of residuals, when the price is lagged for 15 minutes, corresponding to the last-minute bidding case. Panel (b) plots the distribution of residuals, when the price is lagged for 90 minutes, corresponding to the 75-minute bid lead time case.

**Table A1: Variation in Submitted Bid Quantities**

|                                   | Bid Quantity (MW) |       |       |       |
|-----------------------------------|-------------------|-------|-------|-------|
|                                   | (1)               | (2)   | (3)   | (4)   |
| Adjusted $R^2$                    | 0.794             | 0.866 | 0.812 | 0.023 |
| Battery fixed effects             | ✓                 |       |       |       |
| Battery-by-day fixed effects      |                   | ✓     |       |       |
| Battery-by-interval fixed effects |                   |       | ✓     |       |
| Day-by-interval fixed effects     |                   |       |       | ✓     |

Notes: This table explores the data variation in submitted bid quantities using bids information for each battery at 15-minute frequency in 2024. We control for battery fixed effects in column (1), battery-by-day fixed effects in column (2), battery-by-interval-of-day fixed effects in column (3), and day-by-interval fixed effects in column (4). We present the adjusted  $R^2$  in the table.

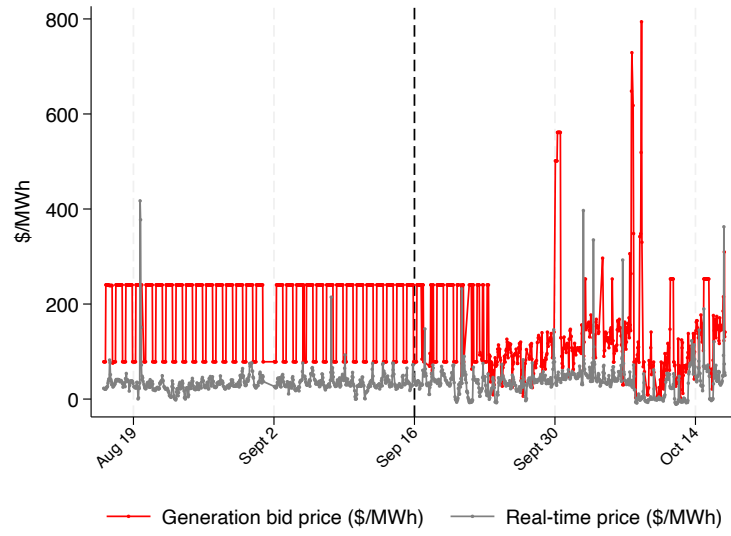
## B Automated bidding adoption

We present a case study of automated bidding adoption in ERCOT, focusing on Flat Top (FLT), a one-hour battery with a nameplate capacity of 10 MW operated by Key Capture Energy.<sup>19</sup> We focus on a sample spanning from August 16 to October 16, 2021. We begin by examining the evolution of average bid prices for FLT at the 15-minute level over this period. As shown in Figure A11, the variation in average bid prices increases substantially following the adoption of automated bidding.

We then compare hourly average bid prices and awarded quantities with the hourly pattern of real-time market prices, before and after September 16, 2021. Figure A12 shows that, prior to automation, bid prices take only two discrete values throughout the day, and awarded quantities are weakly aligned with real-time market prices. After automation, bid prices exhibit much richer intraday variation, and awarded quantities track real-time market prices more closely.

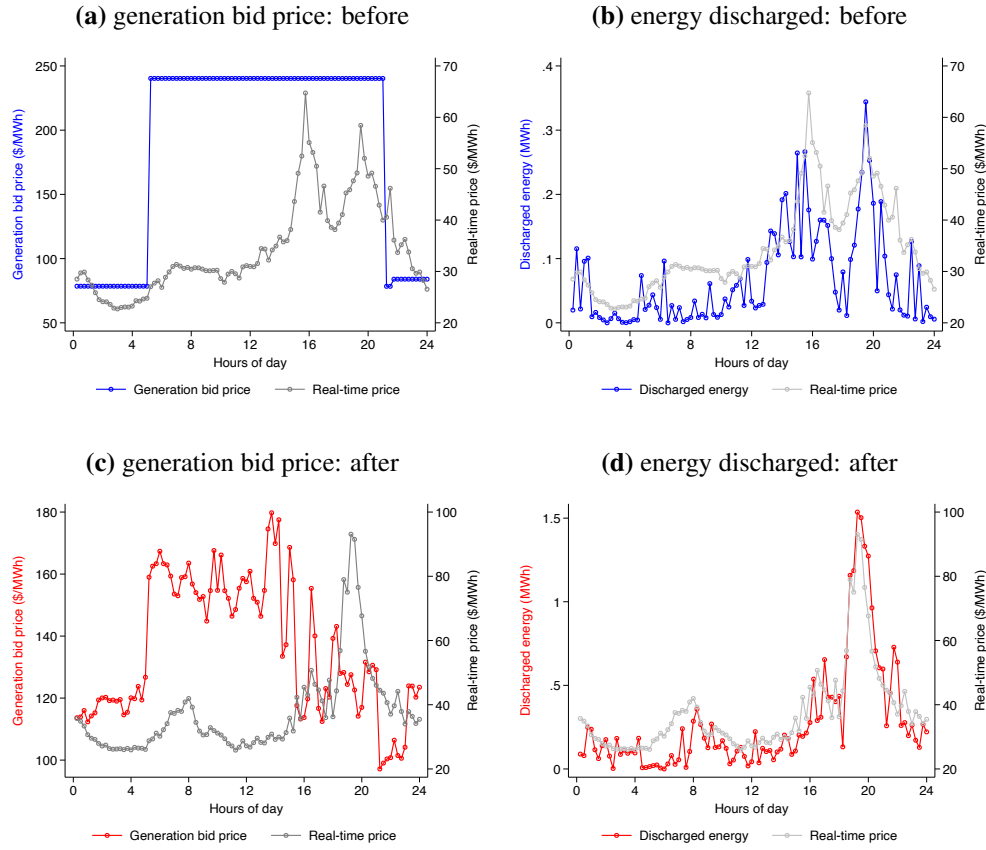
<sup>19</sup>Another battery, Worsham (WRS), also a one-hour, 10 MW unit operated by the same firm, exhibits similar changes in bidding and operational patterns following the adoption of automated bidding on September 16, 2021. To streamline the presentation, we suppress the Worsham results in this draft. We thank Danny Musher of Key Capture Energy for sharing this institutional detail.

**Figure A11:** Automated Bidding Adoption by Flat Top



Notes: This figure plots the daily generation bid prices and real-time electricity prices of Flat Top, where the vertical dashed line is the time it adopted automated bidding (September 16, 2021). Panels (b) and (c) plot the generation bid prices and energy discharged by hour of day before the automated bidding adoption, and Panels (d) and (e) plot the generation bid prices and energy discharged by hour of day after the automated bidding adoption.

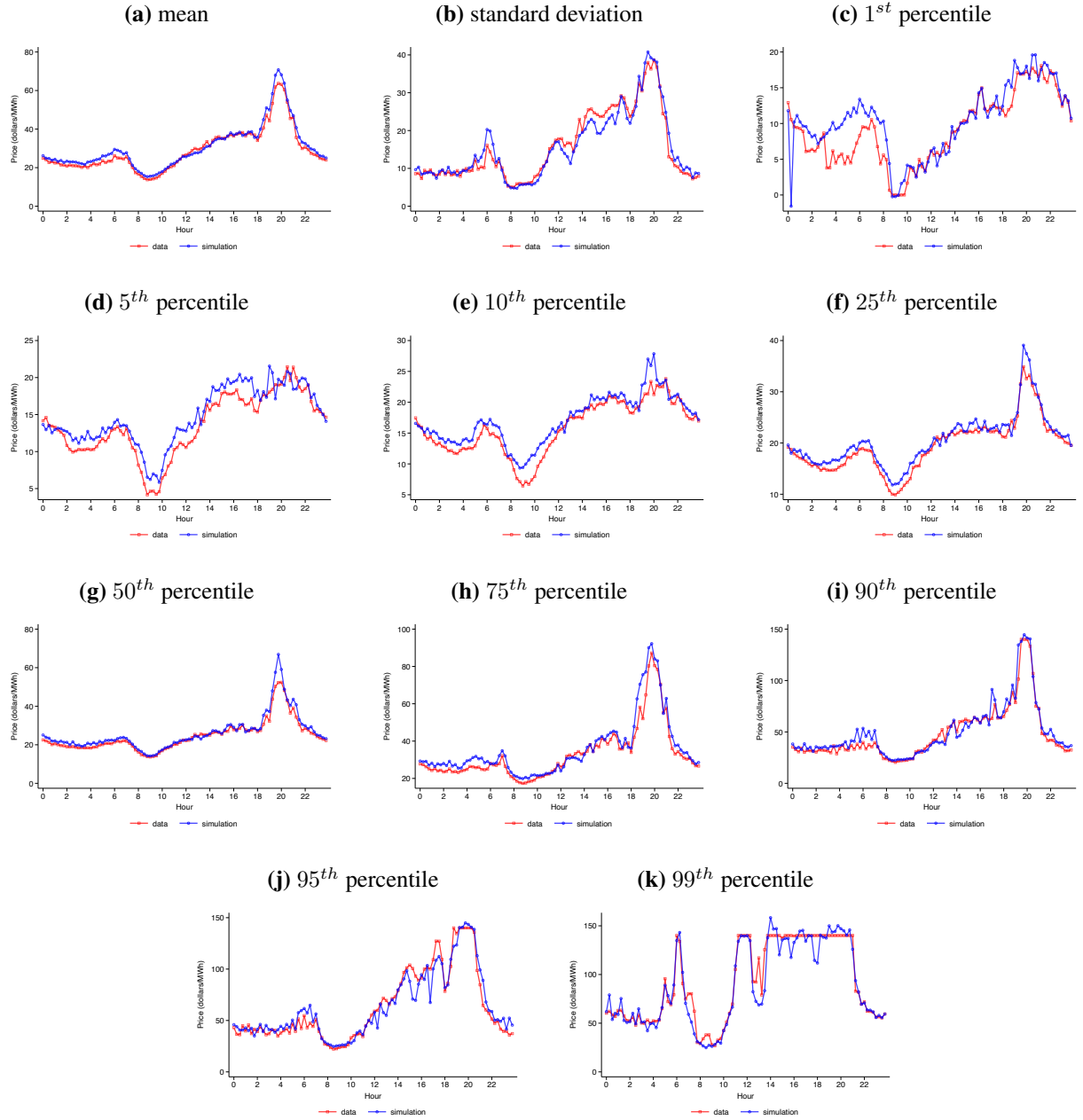
**Figure A12: Automated Bidding Adoption by Flat Top**



Notes: This figure plots the bidding and operation of the battery Flat Top from August to October 2021, before and after it adopted automated bidding softwares. Panels (a) and (b) plot the generation bid prices and energy discharged by hour of day before the automated bidding adoption, and Panels (c) and (d) plot the generation bid prices and energy discharged by hour of day after the automated bidding adoption.

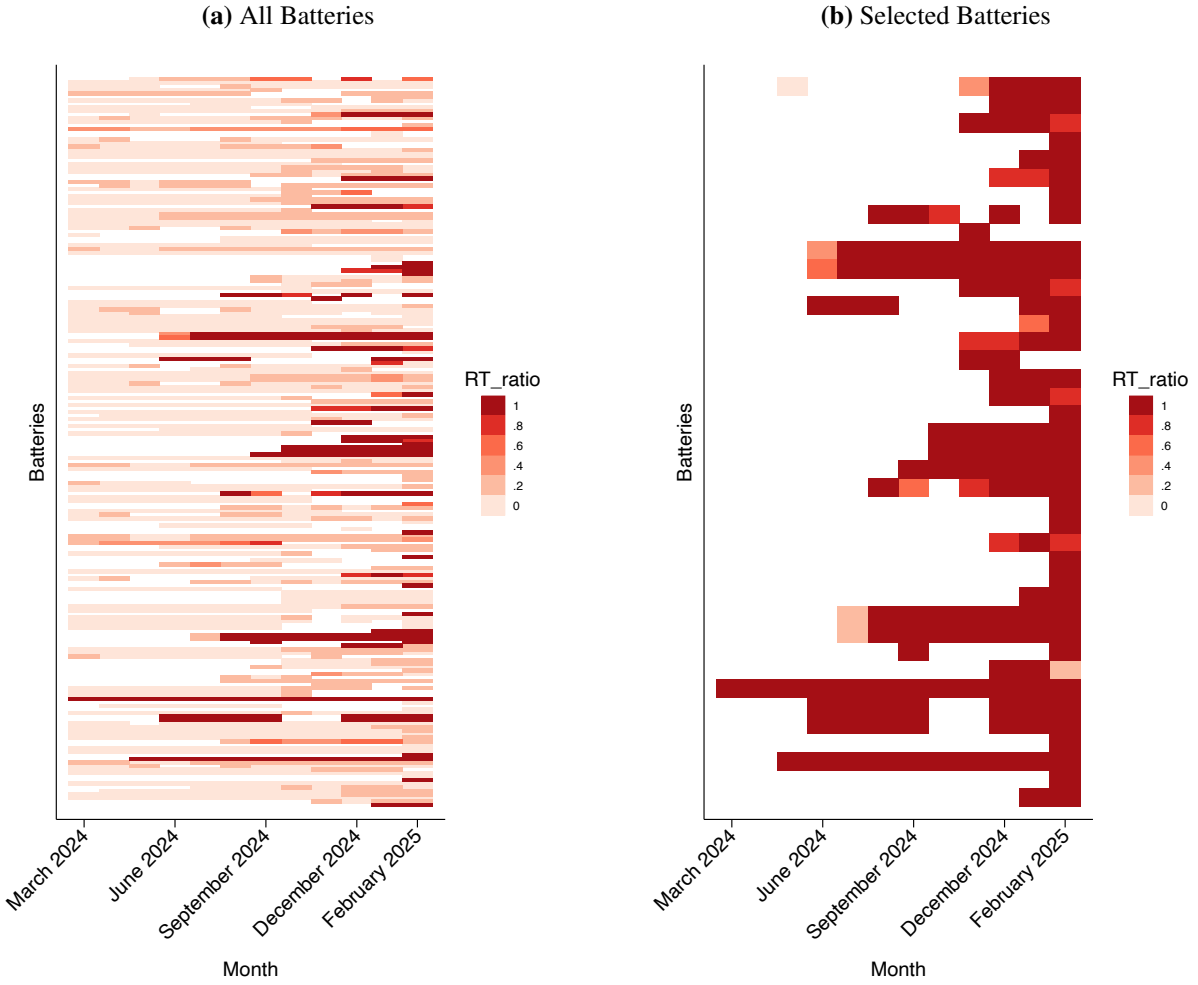
## C Details on model estimation

**Figure A13: Key Moments of Simulated Prices and Data**



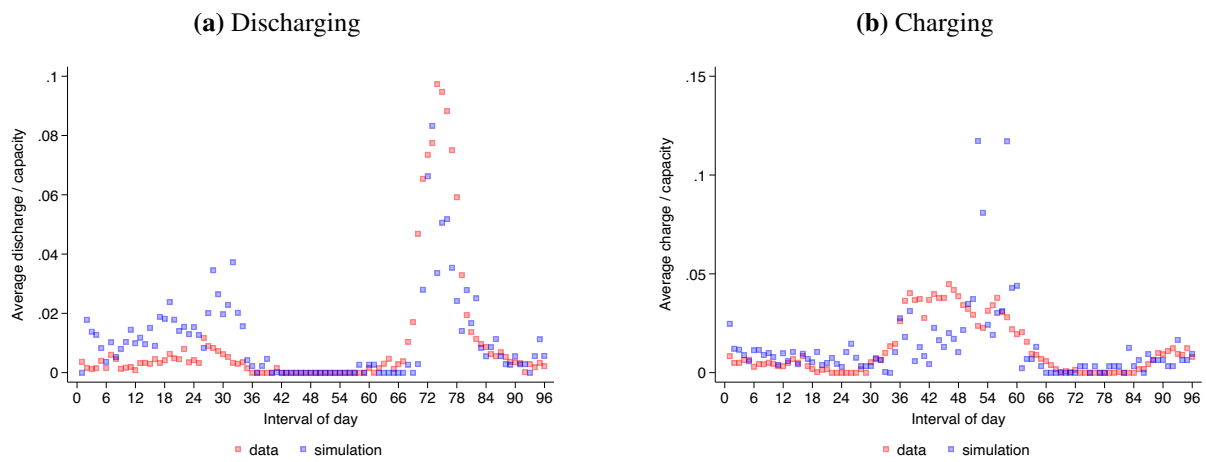
Notes: This figure compares the means (Panel (a)), standard deviations (Panel (b)), and a set of percentiles (Panel (c) to (k)) of simulated prices and realized prices at 15-minute frequency. The price data is from December 2024 to February 2025 at the settlement point where Casl Gap Battery sits. The statistics of simulated prices are denoted in red squares, and the statistics of realized price data are represented by blue circles.

**Figure A14: Batteries Exclusive on RTM: March 2024 - February 2025**



Notes: This figure plots the batteries' share of energy discharged into the real-time energy market in the total sum of both real-time energy market and day-ahead ancillary service market. Panel (a) plots the monthly shares for all active batteries in ERCOT from March 2024 to February 2025. Panel (b) plots the monthly shares for 23 batteries which has at least one season of exclusive real-time market operation between March 2024 and February 2025.

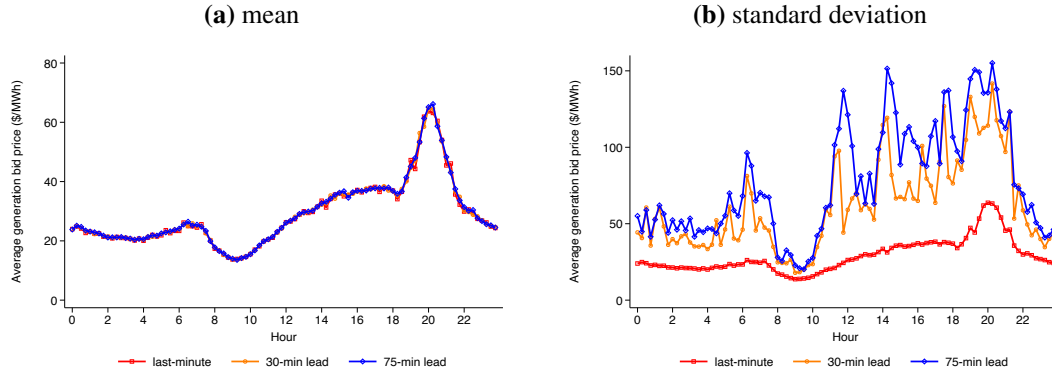
**Figure A15:** Comparison between model-predicted charging/discharging and data



Notes: This figure compares the model-predicted discharging pattern (Panel (a)) and charging pattern (Panel (b)) with data. The simulation is for Casl Gap in the summer of 2024.

## D Details on model simulation

**Figure A16: Mean and Standard Deviations of Price Projections under Lead Time**



Notes: This figure plots the means (Panel (a)) and standard deviations (Panel (b)) of projected prices under various lead time, conditional on the average prices in the starting interval. Red squares represent means and standard deviations of projected prices conditional on average prices 15 minutes ago (last-minute bidding). Orange circles represent means and standard deviations of projected prices conditional on average prices 45 minutes ago (with 30-minute lead time). Blue diamonds represent means and standard deviations of projected prices conditional on average prices 90 minutes ago (with 75-minute lead time).